

Sex Stratified Relationships of Tuberculosis Prevalence and Notification in Low- and Middle-income Countries: A Systematic Review and Meta-analysis

2025-05-07

ABOUT THIS SCRIPT

This script reads in the clean dataset from data folder and then runs fits a Bayesian generalized multivariate multilevel model on the studies that report sex stratified prevalence results. This model aims to estimate sex stratified prevalence adjusted for geographic factors.

```
## [1] "distribution of samples for calculation:"
##
## adj.prev100k.ci.bacteriological.tb  adj.prev100k.ci.smear.positive.tb
##                                43                                2
##          n.bacteriological.tb          n.culture.positive.tb
##                                20                                6
##          n.smear.positive.tb                                none
##                                12                                8
## prev100k.ci.bacteriological.tb prev100k.ci.smear.positive.tb
##                                9                                6
## prev100k.smear.positive.tb
##                                1
```

Descriptive statistics

```
## [1] "99 unique surveys with sex-stratified bacteriological positive TB"

## [1] "91 unique publications"

## [1] "93 studies report participant numbers totaling 4016635 participants"

## [1] "45.76% of participants were male"
```

Risk of bias statistics

```
table(cleanSexDF0$study.quality.summary)
```

```
##
##           Low risk of bias - Further research is very unlikely to change our confidence in the estimate
##                                           43
##           Moderate risk of bias - Further research is likely to have an important impact on our confidence in the estimates
##                                           40
##           High risk of bias - Further research is very likely to have an important impact on our confidence in the estimates
##                                           13
##                                           Other: N/A
##                                           1
## Other: High risk of bias - Further research is very likely to have an important impact on our confidence\nin the estimates
##                                           1
##                                           Other: Medium
##                                           1
```

```
## [1] "distribution of samples for calculation:"
##
## adj.prev100k.ci.bacteriological.tb  adj.prev100k.ci.smear.positive.tb
##                                43                                2
##           n.bacteriological.tb           n.culture.positive.tb
##                                20                                6
##           n.smear.positive.tb           none
##                                12                                8
## prev100k.ci.bacteriological.tb  prev100k.ci.smear.positive.tb
##                                9                                6
##           prev100k.smear.positive.tb
##                                1
```

```
## [1] "distribution of samples for calculation:"
##
## adj.prev100k.ci.smear.positive.tb           n.smear.positive.tb
##                                29                                27
##           none           prev100k.ci.smear.positive.tb
##                                43                                7
## prev100k.smear.positive.tb
##                                1
```

Model 1: Main effect model

Model specification

```
mainEffectMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + (1|study.country) + (1|id),
  data = bactModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    # prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd),
```

```

family = gaussian(),
control = list(adapt_delta = 0.99),
chains = 4,
cores = 4,
iter = 4000)

```

```

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##      ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

```

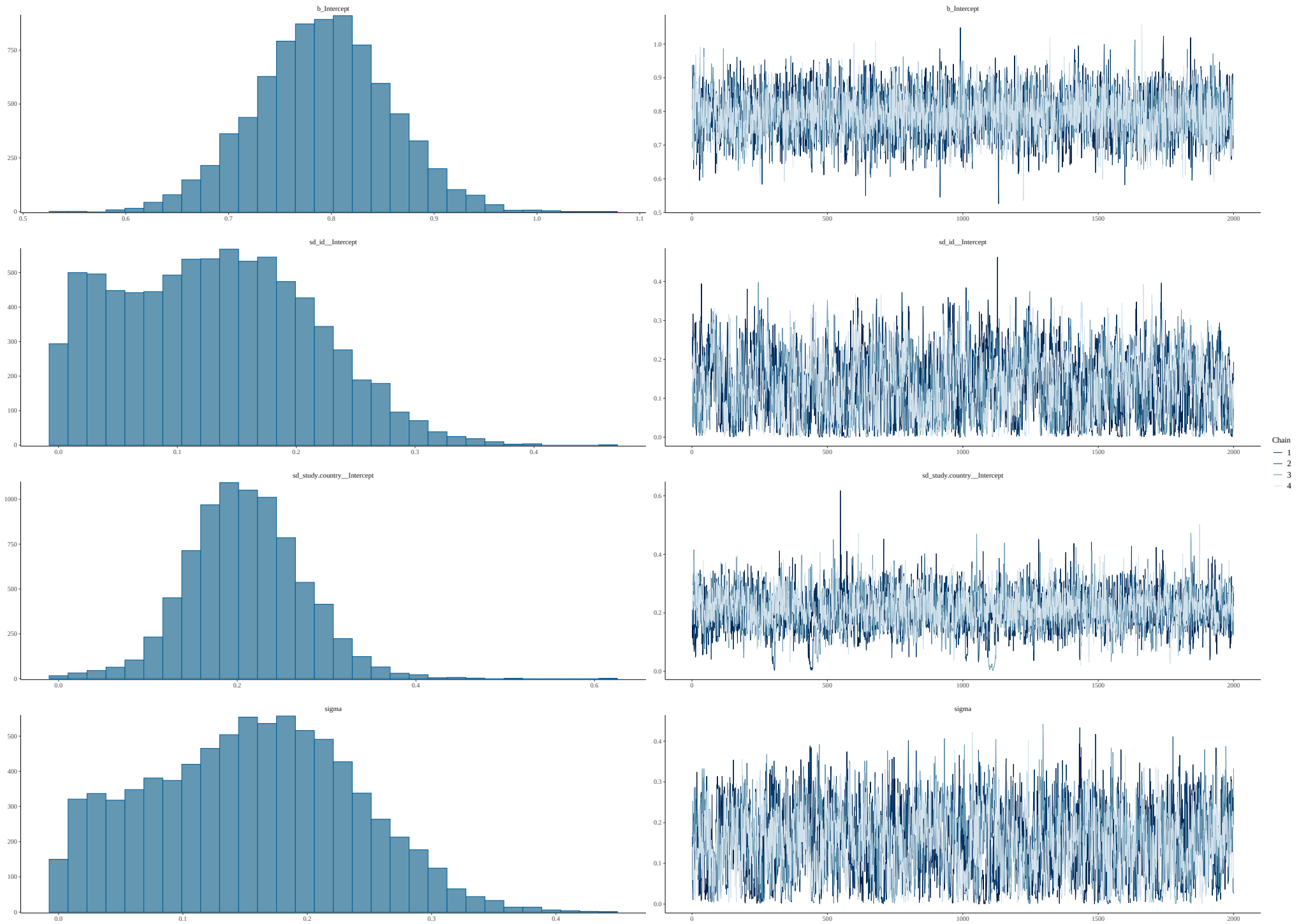
Summarize and check

```

## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + (1 | study.country) + (1 | id)
## Data: bactModelDF (Number of observations: 99)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 99)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.13     0.08    0.00    0.29 1.00     911     2114
##
## ~study.country (Number of levels: 33)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.21     0.06    0.09    0.34 1.00    1985     1614
##
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    0.79     0.07    0.66    0.92 1.00     4863     5628
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma    0.15     0.08    0.01    0.31 1.00     1096     2092
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS

```

and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).



Prediction

Main estimate

```
mainEffectModEst %>% ungroup %>% mean_qi(mfRatio)
```

```
## # A tibble: 1 x 6
##   mfRatio .lower .upper .width .point .interval
##   <dbl>   <dbl>   <dbl>   <dbl> <chr>   <chr>
## 1     2.21     1.94     2.50     0.95 mean    qi
```

Model 2: Effect by world region model

Model specification

```
regionEffectMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + (1|WHO.region) + (1|id),
  data = bactModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    # prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd),
  family = gaussian(),
  control = list(adapt_delta = 0.99),
  chains = 4,
  cores = 4,
  iter = 4000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##           ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

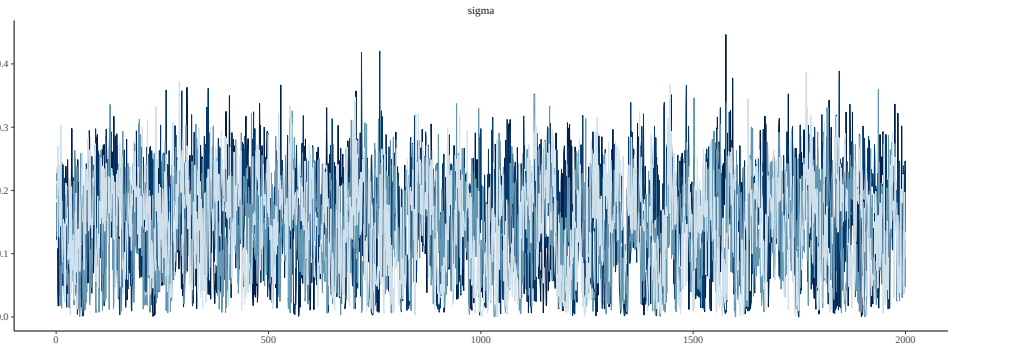
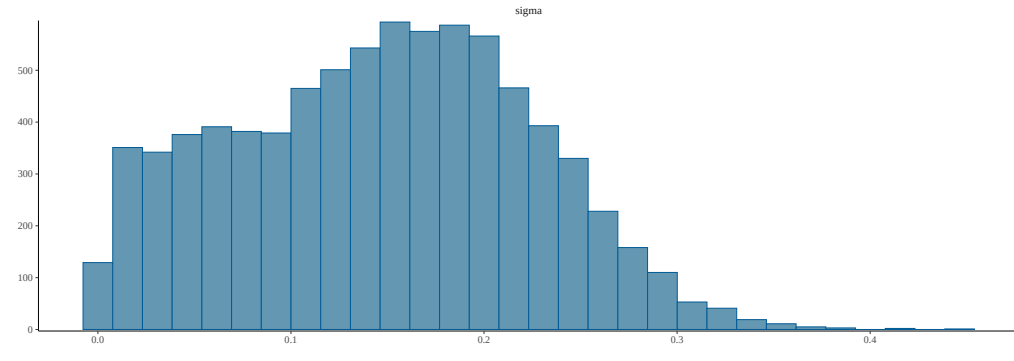
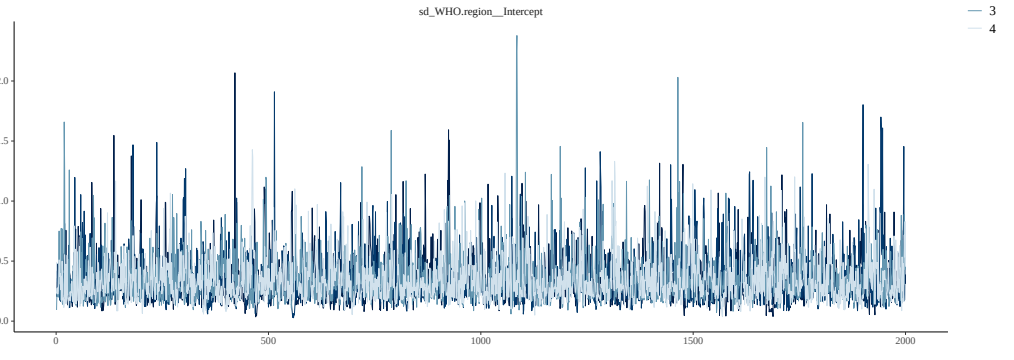
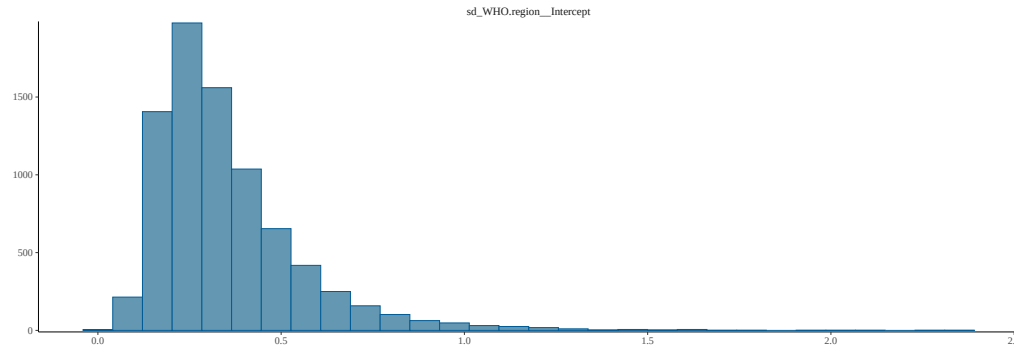
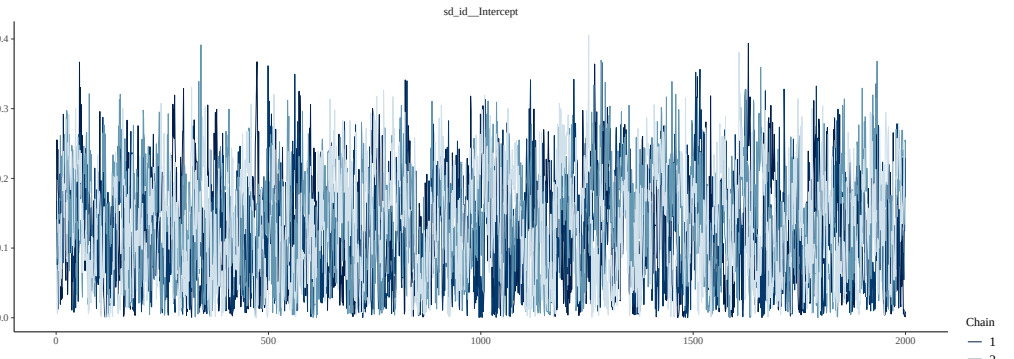
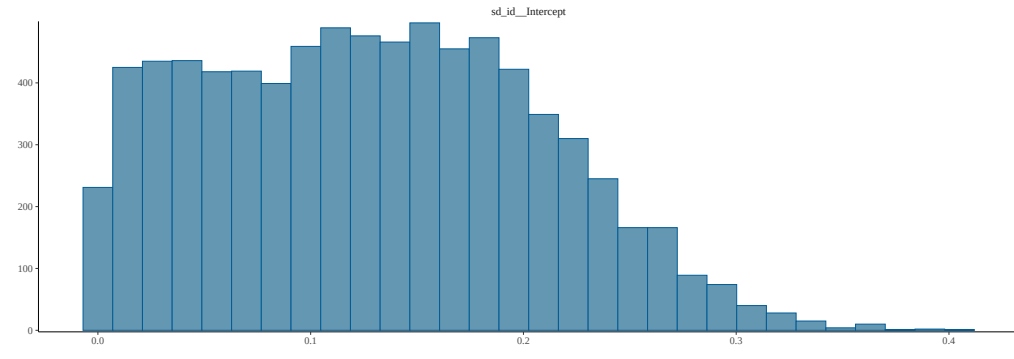
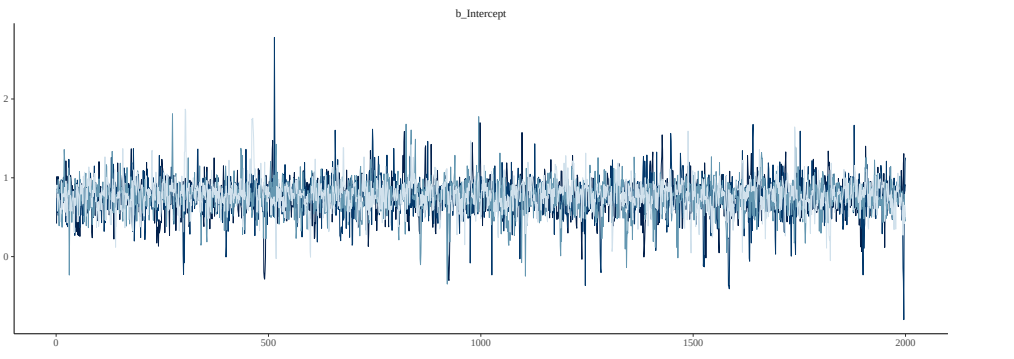
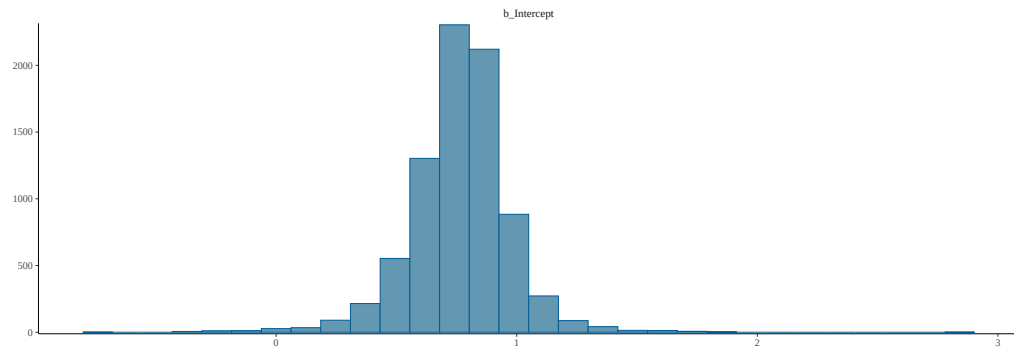
Summarize and check

```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + (1 | WHO.region) + (1 | id)
```

```

## Data: bactModelDF (Number of observations: 99)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 99)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 0.13 0.08 0.01 0.28 1.00 1070 3009
##
## ~WHO.region (Number of levels: 5)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 0.35 0.20 0.12 0.87 1.00 3260 3739
##
## Regression Coefficients:
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept 0.77 0.21 0.32 1.15 1.00 3656 3294
##
## Further Distributional Parameters:
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma 0.15 0.08 0.01 0.29 1.00 1333 2895
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

```



Chain
— 1
— 2
— 3
— 4

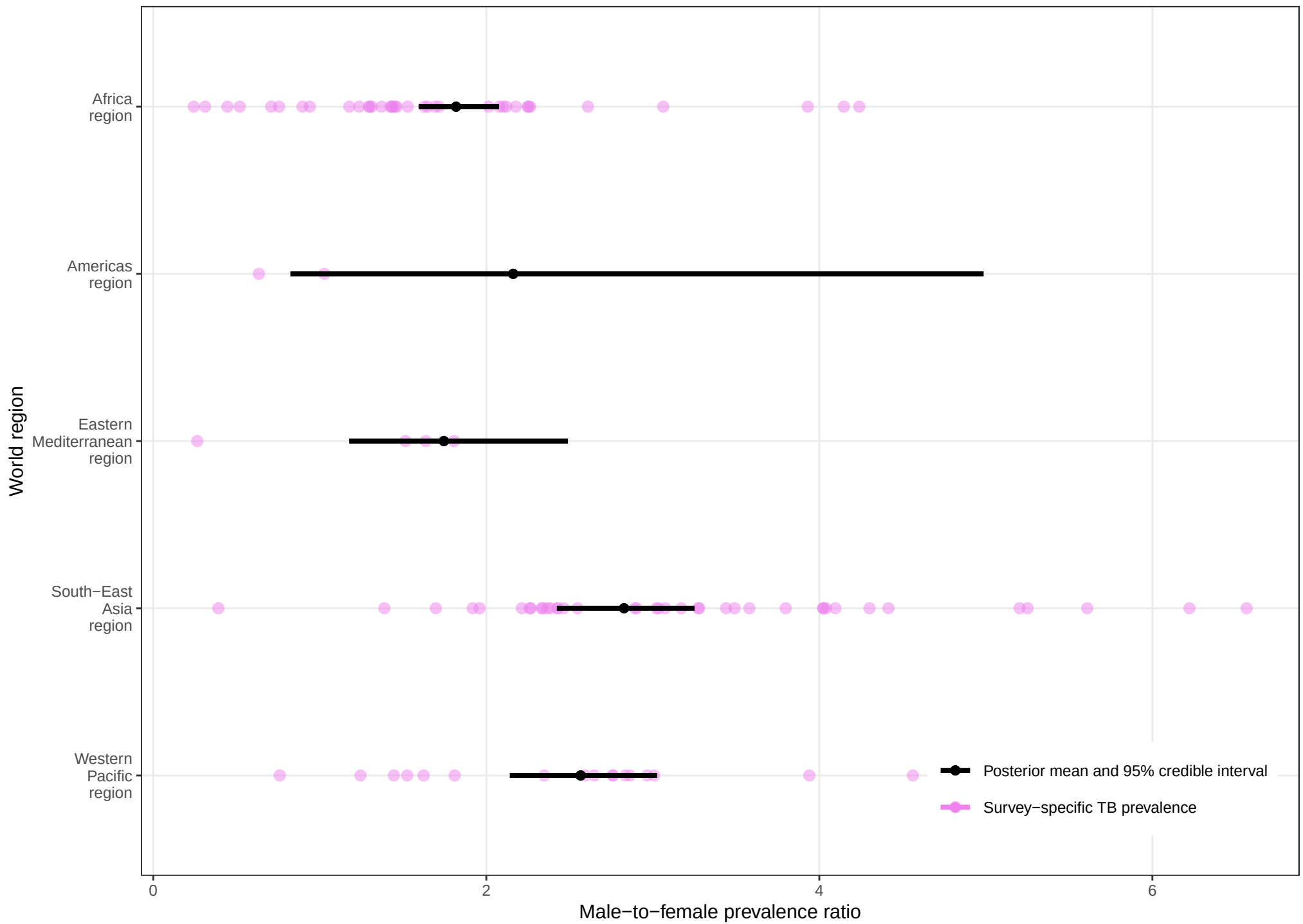
Prediction

Main estimate

```
regionEffectModEst %>% group_by(WHO.region) %>% mean_qi(mfRatio)
```

A tibble: 5 x 7

WHO.region	mfRatio	.lower	.upper	.width	.point	.interval
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1 Africa region	1.82	1.59	2.08	0.95	mean	qi
2 Americas region	2.33	0.823	4.99	0.95	mean	qi
3 Eastern Mediterranean region	1.76	1.18	2.49	0.95	mean	qi
4 South-East Asia region	2.83	2.42	3.25	0.95	mean	qi
5 Western Pacific region	2.57	2.14	3.03	0.95	mean	qi



Model 3: Effect over time

Model specification

```
timeEffectMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + year_z + (1 + year_z | study.country) + (1 | id),
  data = bactModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd) +
    prior(lkj(4), class=cor),
  family = gaussian(),
  control = list(adapt_delta = 0.99),
  chains = 4,
  cores = 4,
  iter = 4000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##      ~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

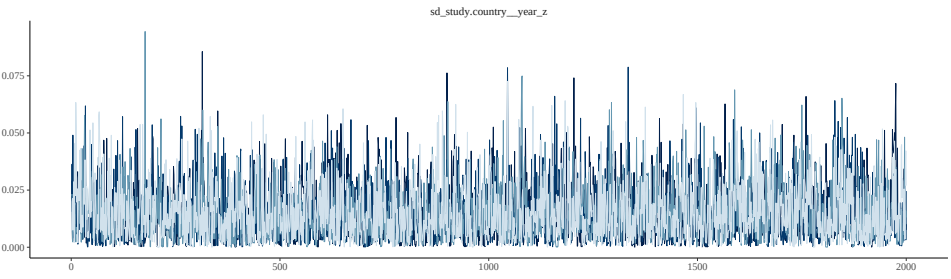
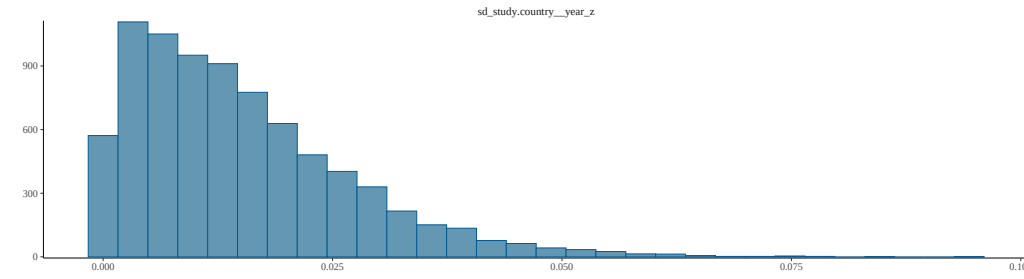
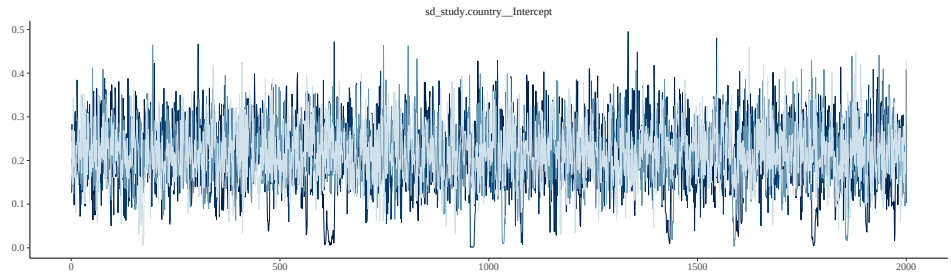
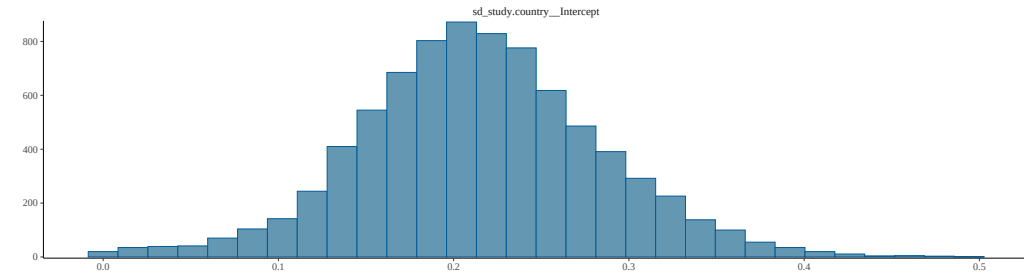
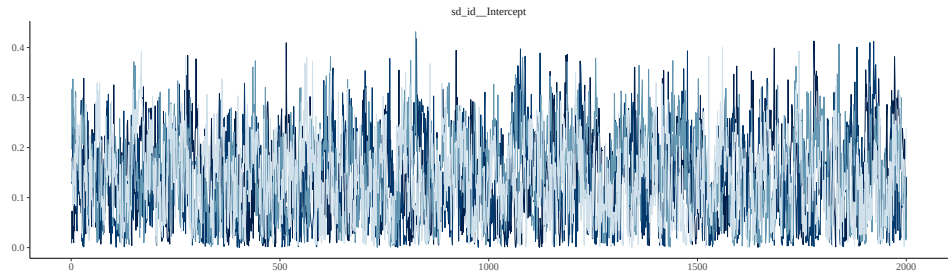
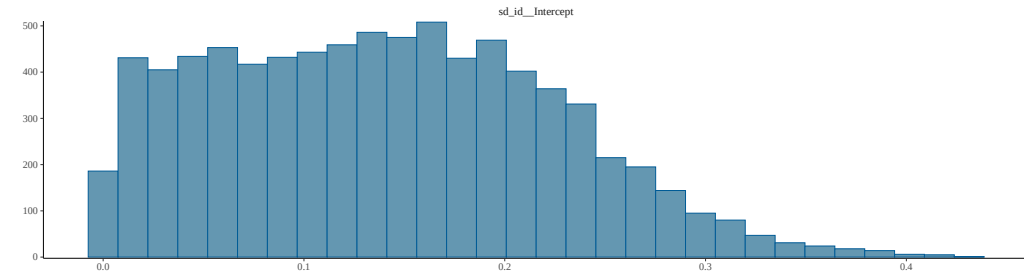
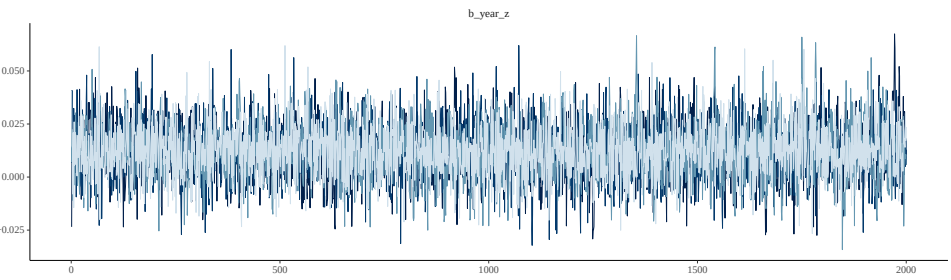
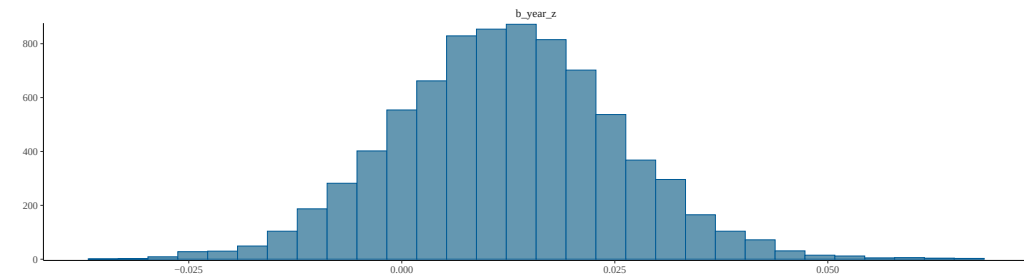
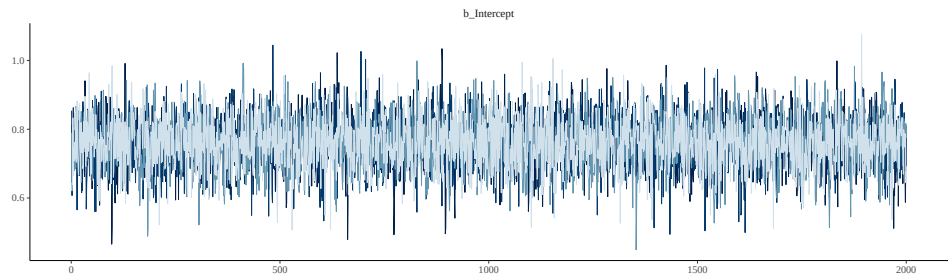
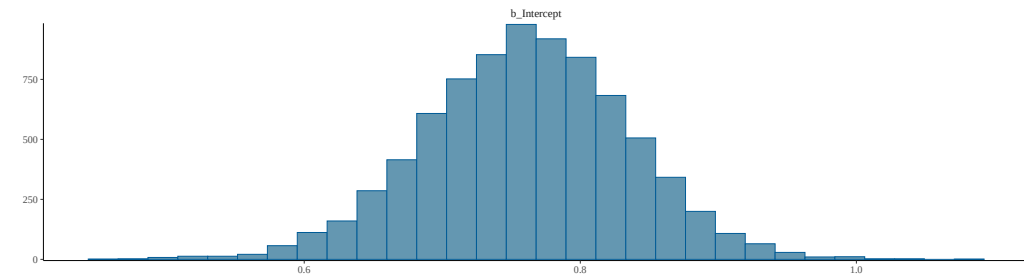
Summarize and check

```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + year_z + (1 + year_z | study.country) + (1 | id)
## Data: bactModelDF (Number of observations: 94)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 94)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 0.14 0.08 0.01 0.31 1.01 814 2407
##
## ~study.country (Number of levels: 33)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
```

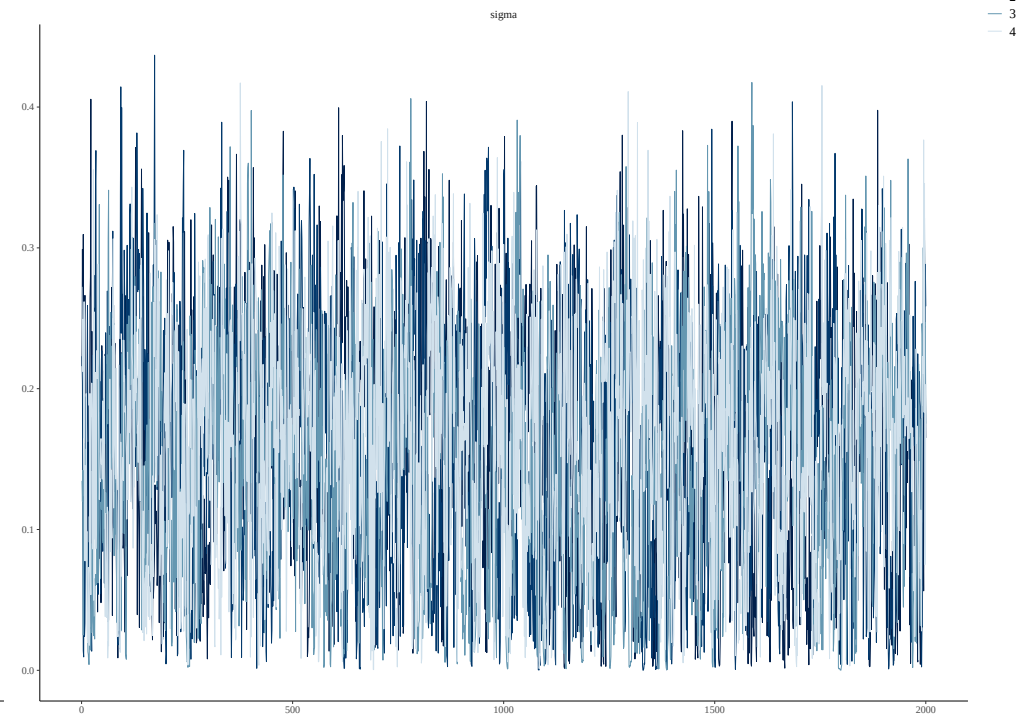
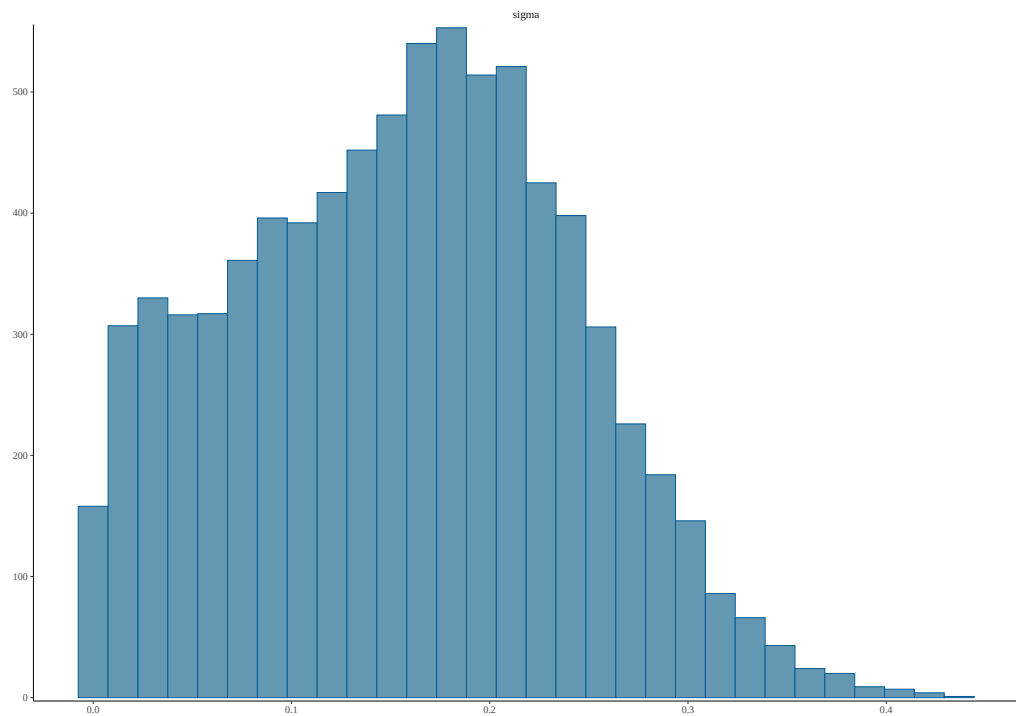
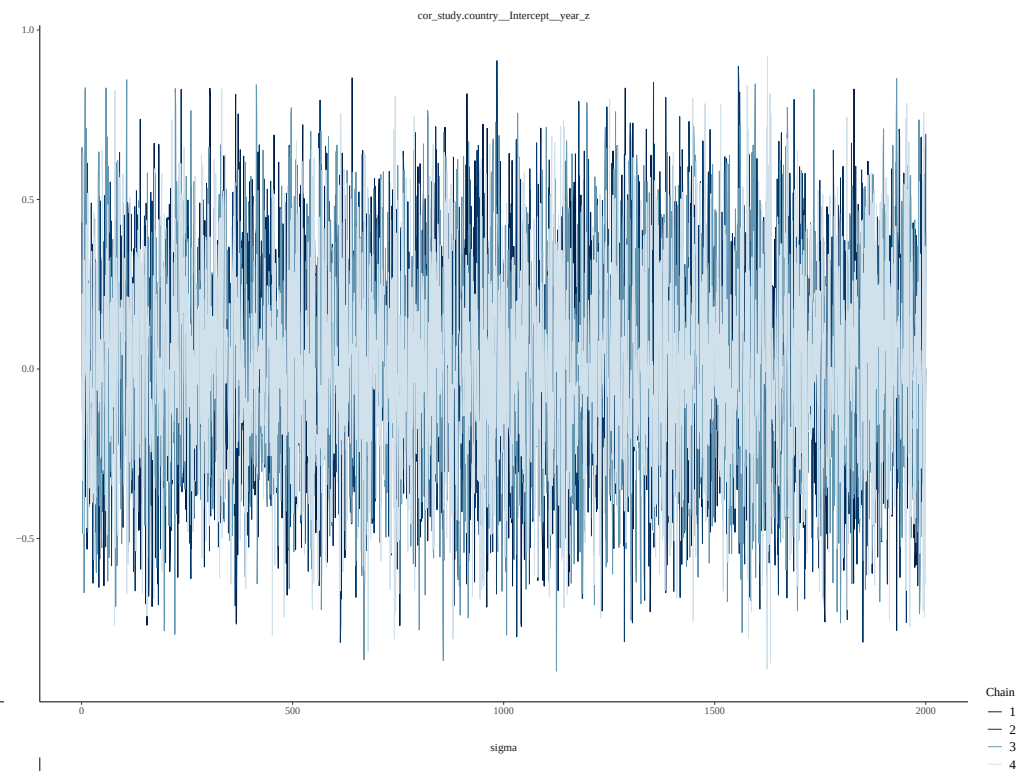
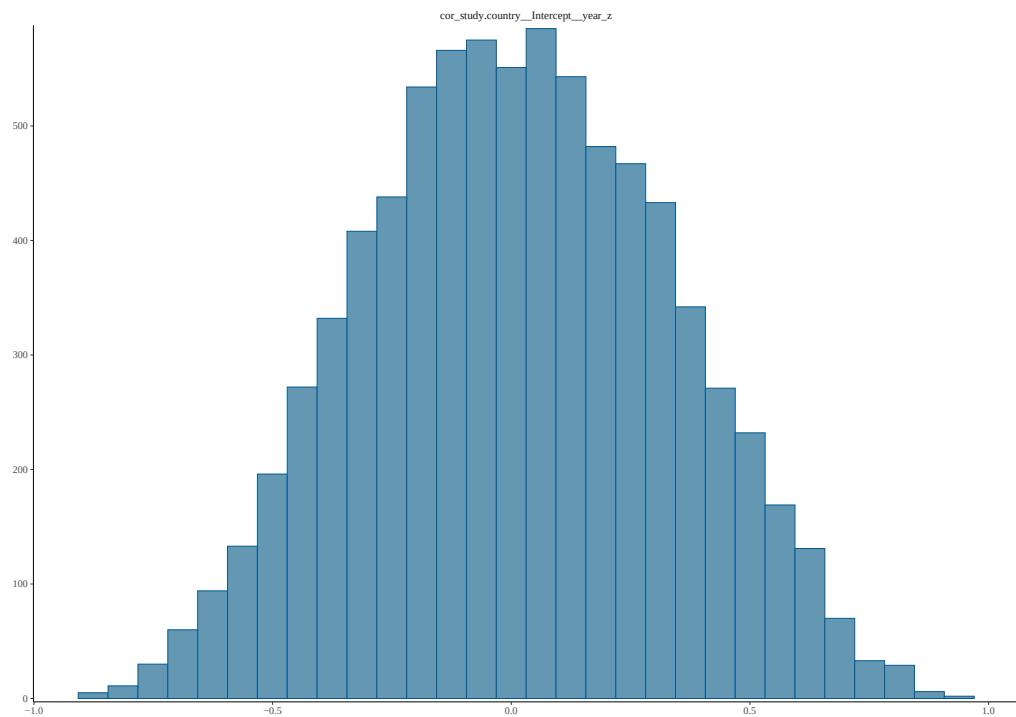
```

## sd(Intercept)          0.21      0.07      0.08      0.35 1.00      2329
## sd(year_z)             0.01      0.01      0.00      0.04 1.00      3138
## cor(Intercept,year_z)  0.01      0.32     -0.59      0.63 1.00      8795
##                               Tail_ESS
## sd(Intercept)          1430
## sd(year_z)             3903
## cor(Intercept,year_z)  6006
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      0.76      0.07      0.61      0.90 1.00      5325      5335
## year_z         0.01      0.01     -0.01      0.04 1.00      5955      5201
##
## Further Distributional Parameters:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma         0.16      0.08      0.01      0.32 1.01      870      2752
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

```



Chain
1
2
3
4



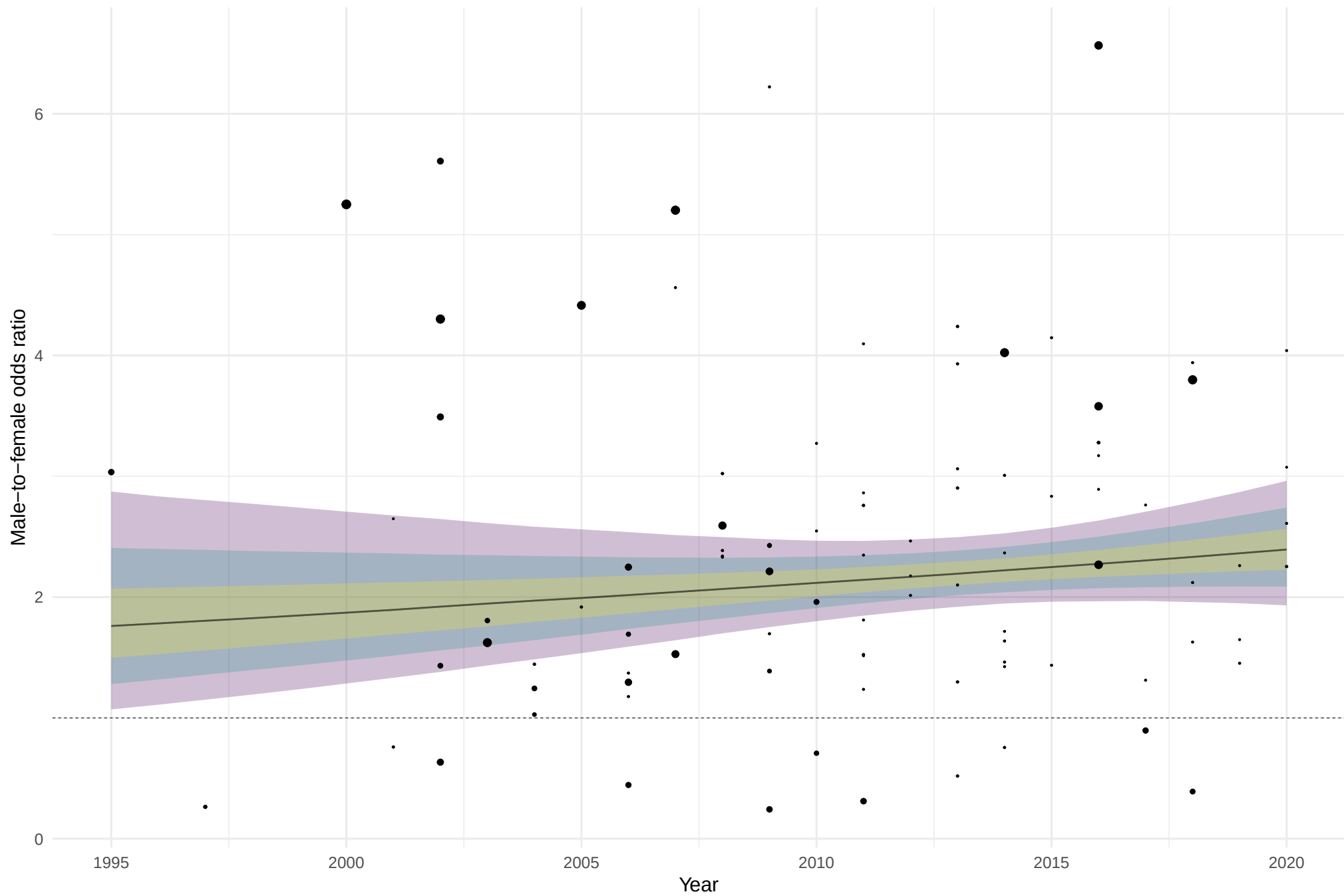
Prediction

Main estimate

```
timeEffectModEst %>%
  group_by(year_z) %>%
  mean_qi(mfRatio) %>%
  mutate(Year = year_z + round(mean(cleanSexDF$study.end.year, na.rm=TRUE))) %>%
  select(Year, mfRatio, .lower, .upper)
```

```
## # A tibble: 26 x 4
```

```
##   Year mfRatio .lower .upper
##   <dbl>   <dbl> <dbl> <dbl>
## 1  1995    1.81  1.07  2.87
## 2  1996    1.83  1.11  2.83
## 3  1997    1.85  1.15  2.80
## 4  1998    1.86  1.19  2.77
## 5  1999    1.88  1.24  2.74
## 6  2000    1.90  1.28  2.71
## 7  2001    1.92  1.33  2.68
## 8  2002    1.94  1.38  2.65
## 9  2003    1.96  1.43  2.61
## 10 2004    1.98  1.48  2.58
## # i 16 more rows
```



Model 4: Region effect over time

Model specification

```
timeEffectRegionMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + year_z * WHO.region + (1 + year_z | study.country) + (1 | id),
  data = bactModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd) +
    prior(lkj(4), class=cor),
  family = gaussian(),
  control = list(adapt_delta = 0.99),
  chains = 4,
  cores = 4,
  iter = 4000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##      ~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

Summarize and check

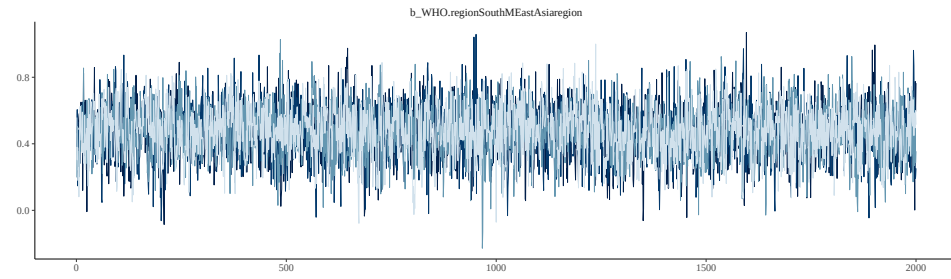
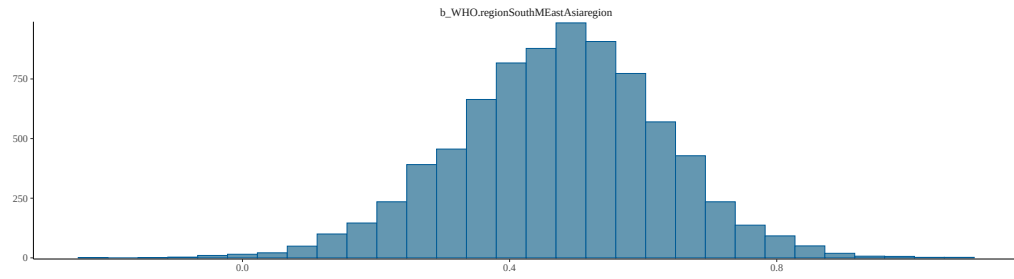
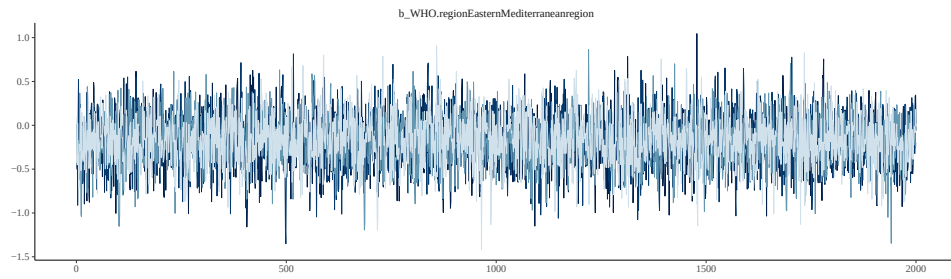
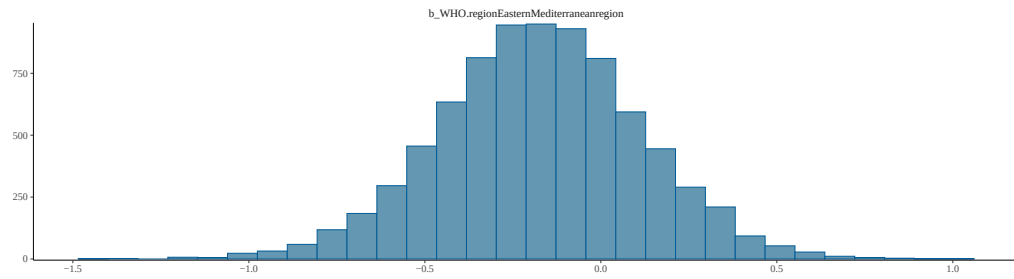
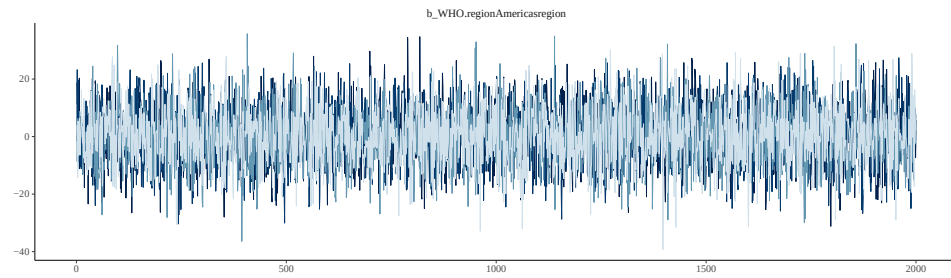
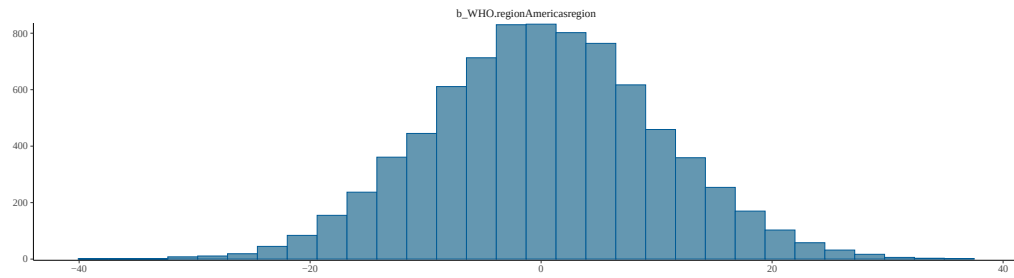
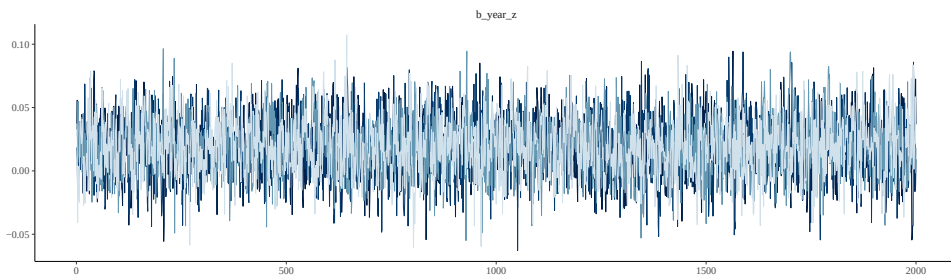
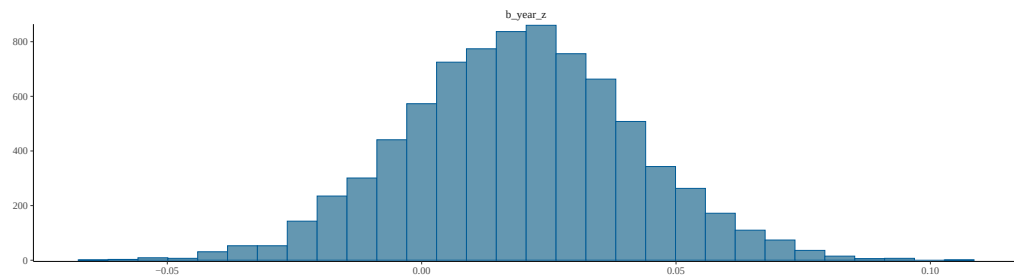
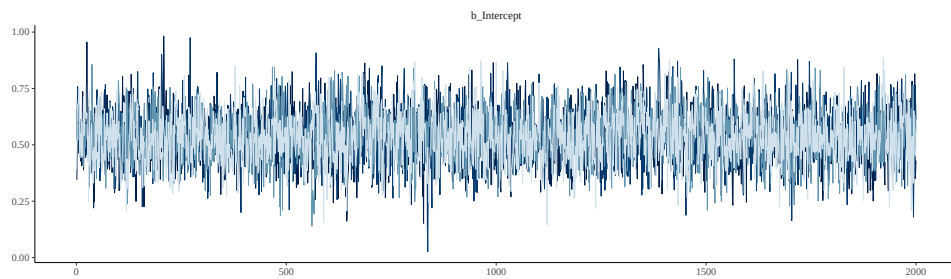
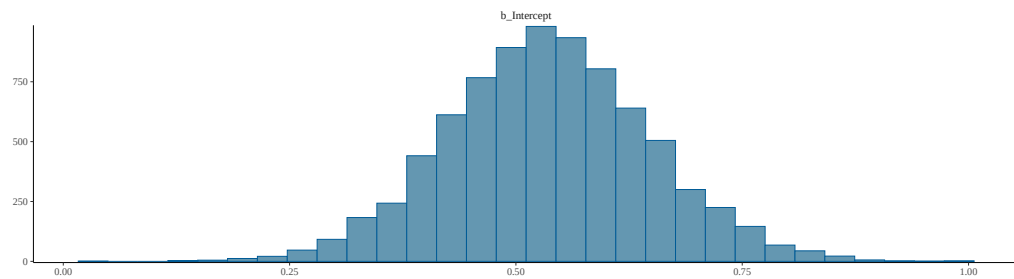
```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + year_z * WHO.region + (1 + year_z | study.country) + (1 | id)
## Data: bactModelDF (Number of observations: 94)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 94)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.14    0.08    0.01    0.30 1.00    866    2565
##
## ~study.country (Number of levels: 33)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
```



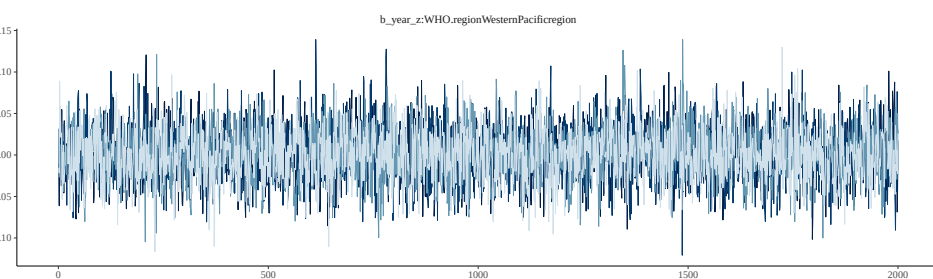
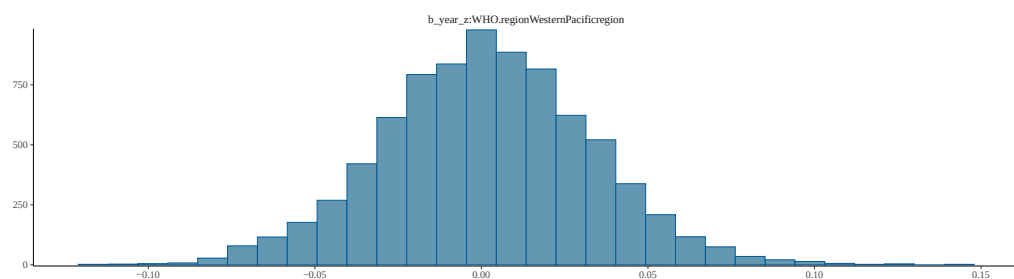
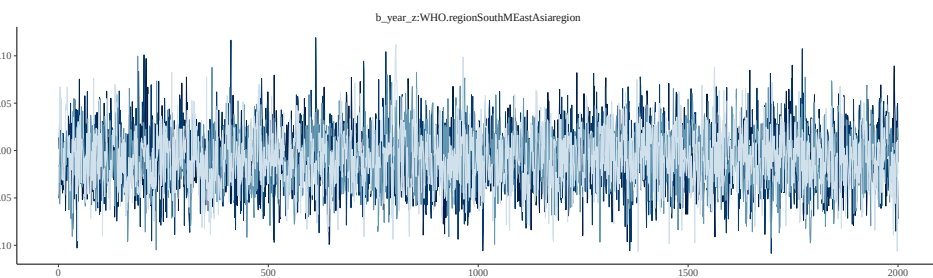
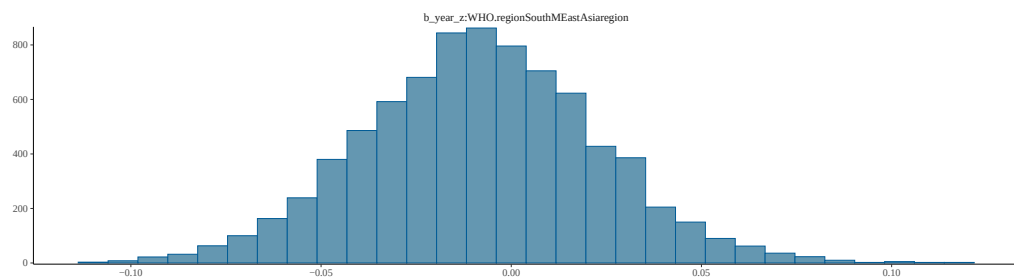
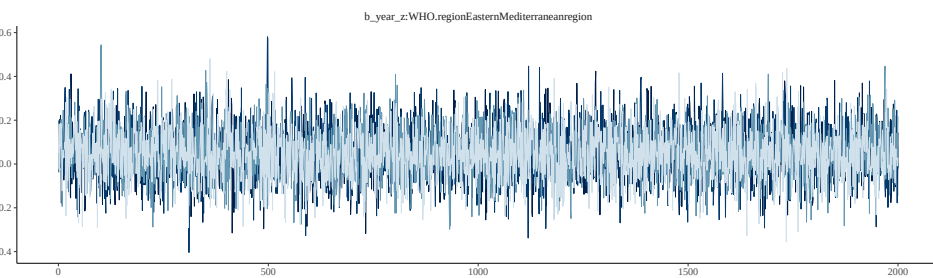
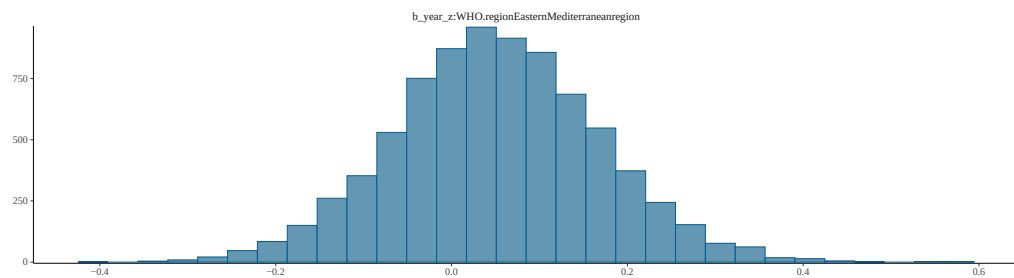
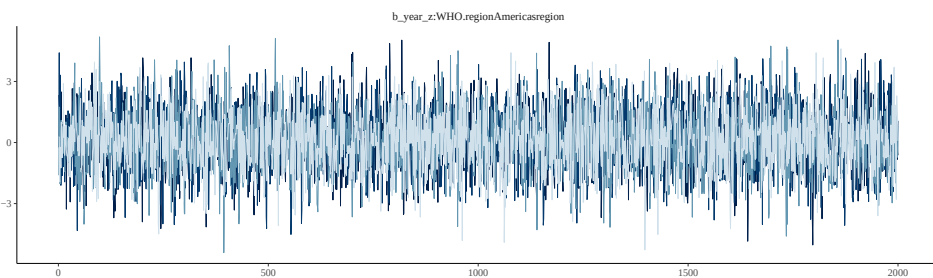
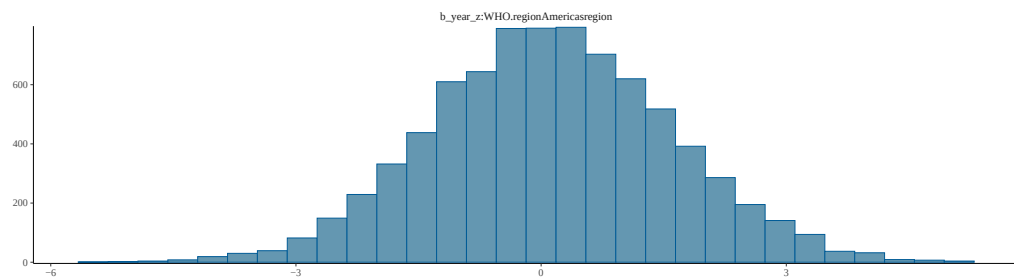
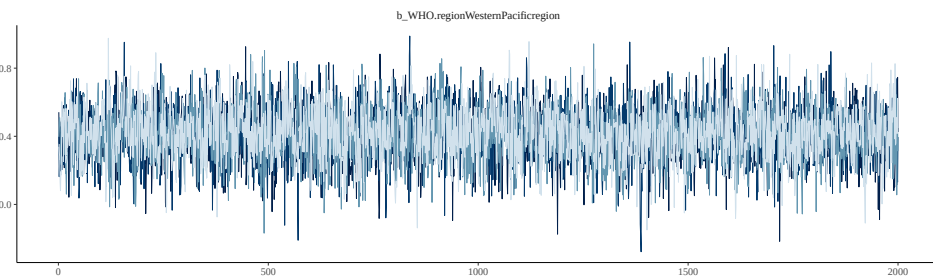
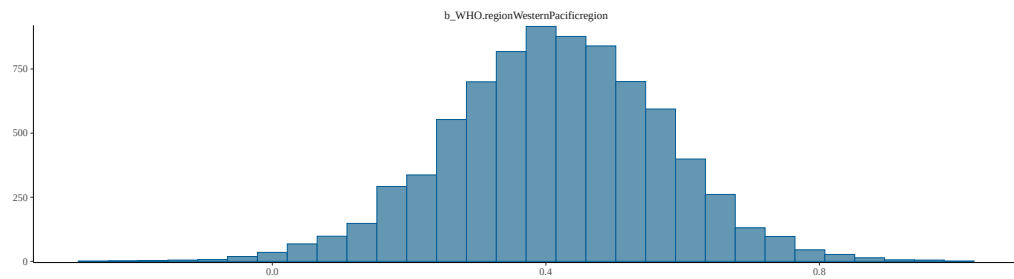
```

## sd(Intercept)          0.09    0.06    0.00    0.24 1.00    2691
## sd(year_z)             0.02    0.01    0.00    0.05 1.00    2856
## cor(Intercept,year_z) -0.01    0.33   -0.63    0.62 1.00   10176
##
##                               Tail_ESS
## sd(Intercept)          4061
## sd(year_z)             3681
## cor(Intercept,year_z)  6151
##
## Regression Coefficients:
##
##               Estimate Est.Error 1-95% CI
## Intercept          0.54      0.11   0.32
## year_z              0.02      0.02  -0.02
## WHO.regionAmericasregion  0.28     9.86 -18.74
## WHO.regionEasternMediterraneanregion -0.18    0.29  -0.75
## WHO.regionSouthMEastAsiaregion  0.47    0.15   0.16
## WHO.regionWesternPacifiregion  0.41    0.16   0.10
## year_z:WHO.regionAmericasregion  0.12    1.47  -2.70
## year_z:WHO.regionEasternMediterraneanregion  0.05    0.12  -0.17
## year_z:WHO.regionSouthMEastAsiaregion -0.01    0.03  -0.07
## year_z:WHO.regionWesternPacifiregion  0.00    0.03  -0.06
##
##               u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          0.76 1.00    4207    5349
## year_z              0.06 1.00    3982    5165
## WHO.regionAmericasregion 19.75 1.00    6954    5998
## WHO.regionEasternMediterraneanregion  0.38 1.00    6047    5530
## WHO.regionSouthMEastAsiaregion  0.77 1.00    4124    4510
## WHO.regionWesternPacifiregion  0.72 1.00    4949    5530
## year_z:WHO.regionAmericasregion  3.04 1.00    6390    5448
## year_z:WHO.regionEasternMediterraneanregion  0.28 1.00    8139    6099
## year_z:WHO.regionSouthMEastAsiaregion  0.05 1.00    4354    5076
## year_z:WHO.regionWesternPacifiregion  0.06 1.00    4616    5285
##
## Further Distributional Parameters:
##
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma    0.16    0.08    0.01    0.31 1.00    966    1700
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

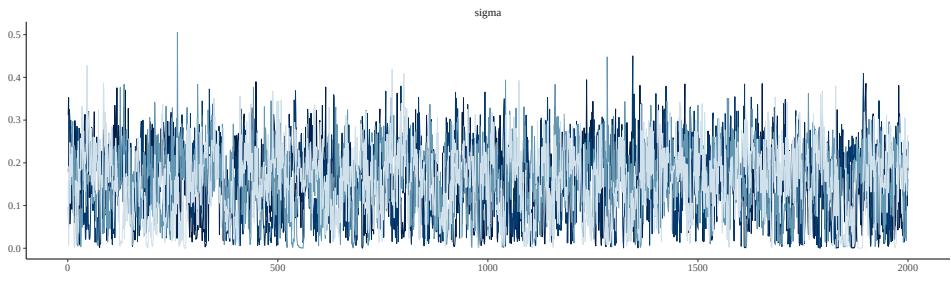
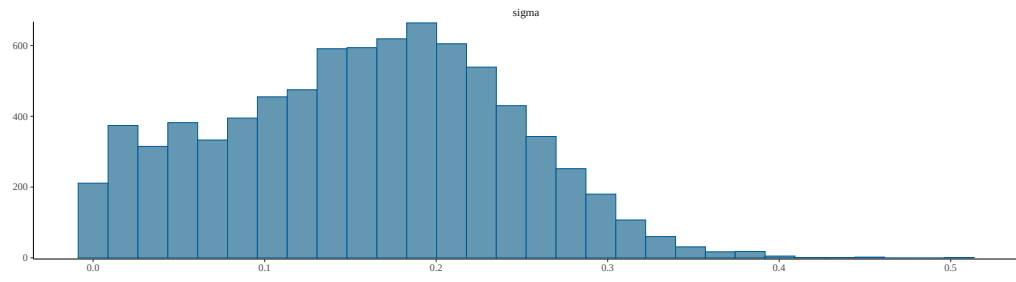
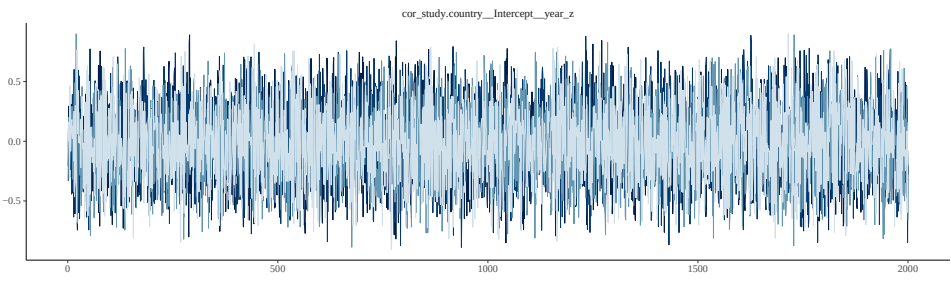
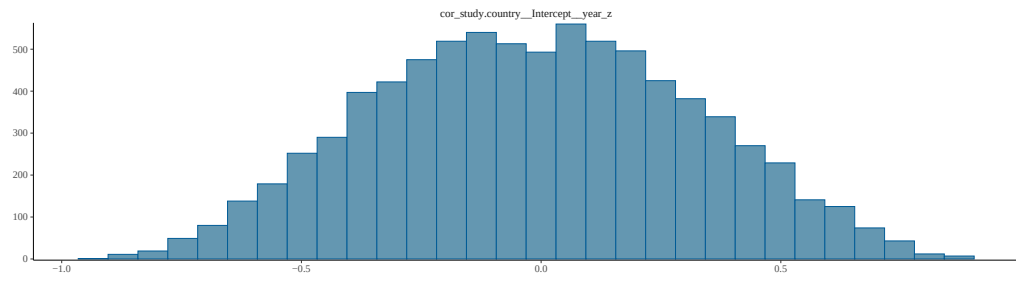
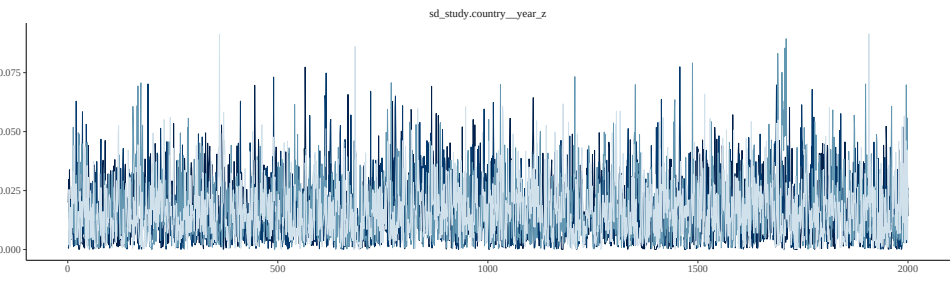
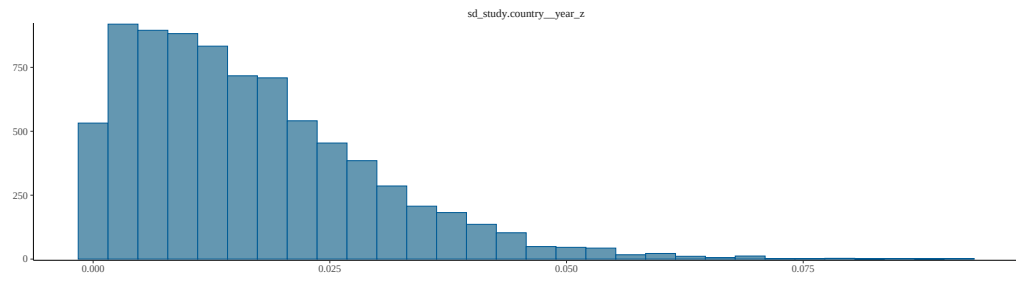
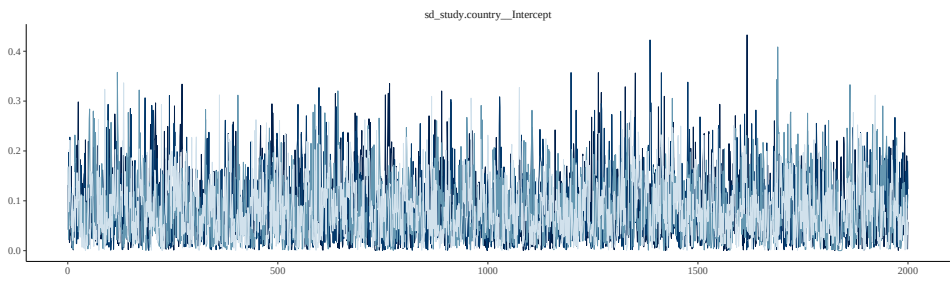
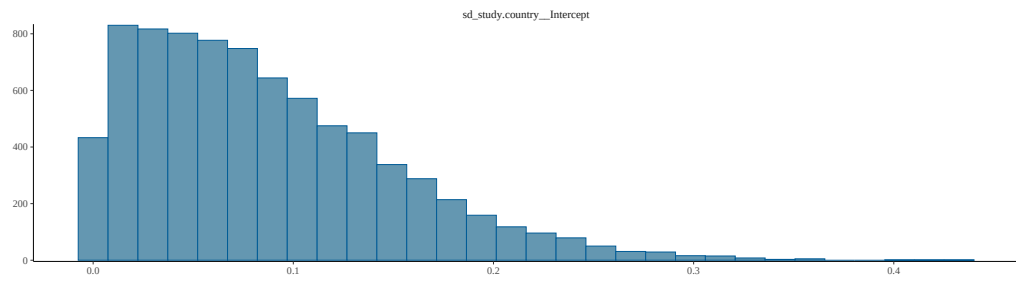
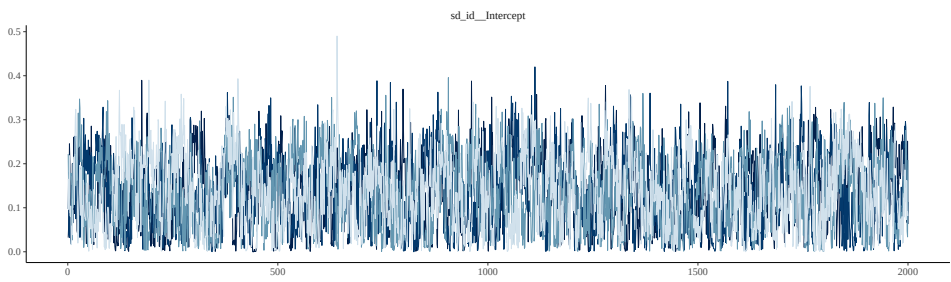
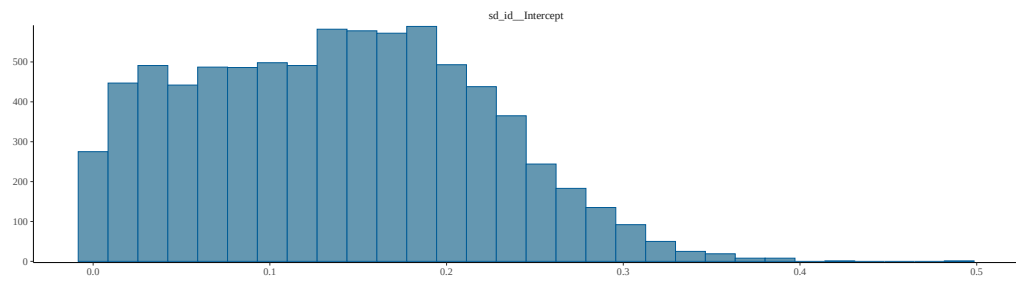
```



Chain
1
2
3
4



Chain
1
2
3
4



Prediction

Main estimate

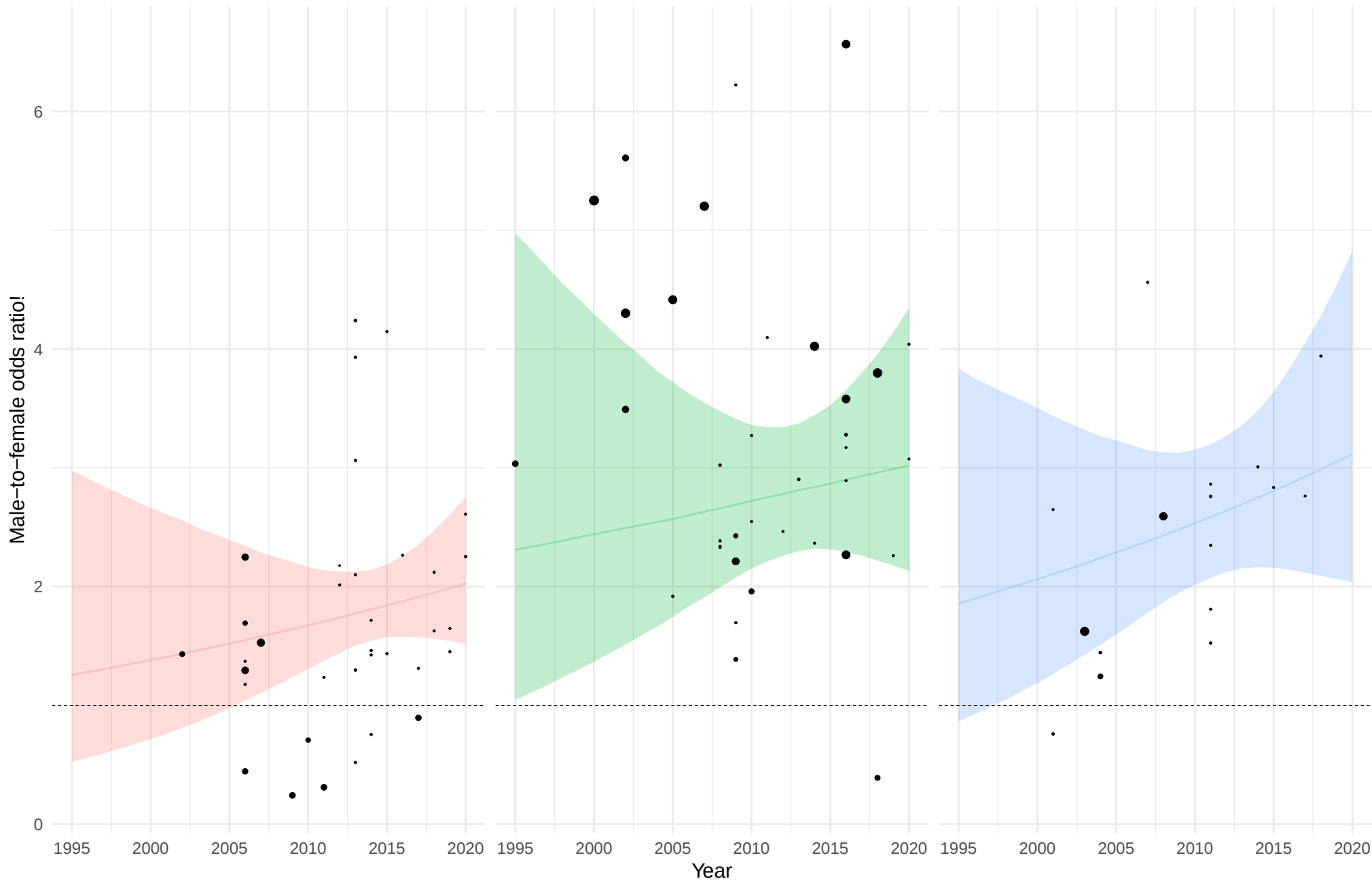
```
timeEffectRegionModEst %>%
  group_by(year_z) %>%
  mean_qi(mfRatio) %>%
  mutate(Year = year_z + round(mean(cleanSexDF$study.end.year, na.rm=TRUE))) %>%
  select(Year, mfRatio, .lower, .upper)
```

```
## # A tibble: 26 x 4
##   Year mfRatio .lower .upper
##   <dbl>   <dbl> <dbl> <dbl>
## 1 1995 1.63e17 0.238 4.93
## 2 1996 1.24e15 0.268 4.77
## 3 1997 9.69e12 0.298 4.60
## 4 1998 7.84e10 0.333 4.45
## 5 1999 6.60e 8 0.372 4.30
## 6 2000 5.95e 6 0.416 4.15
## 7 2001 6.12e 4 0.465 4.02
## 8 2002 8.63e 2 0.522 3.90
## 9 2003 3.20e 1 0.585 3.79
## 10 2004 7.47e 0 0.653 3.68
## # i 16 more rows
```

Africa region

South-East Asia region

Western Pacific region



Univariate analysis of other covariates

Model 6: National versus subnational

Model specification

```
natEffectMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + natRep + (1 | study.country) + (1 | id),
  data = bactModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd),
  family = gaussian(),
  control = list(adapt_delta = 0.99),
  chains = 4,
  cores = 4,
  iter = 4000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##      ~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

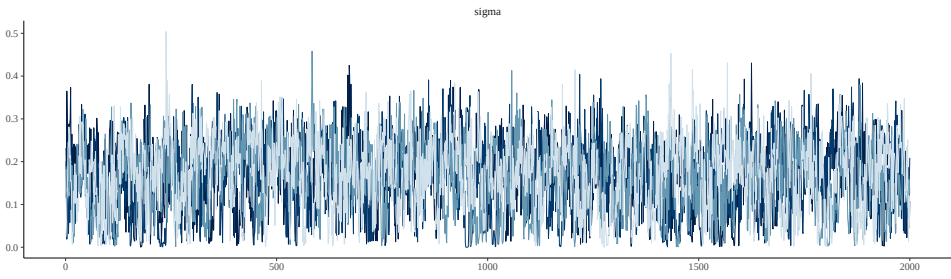
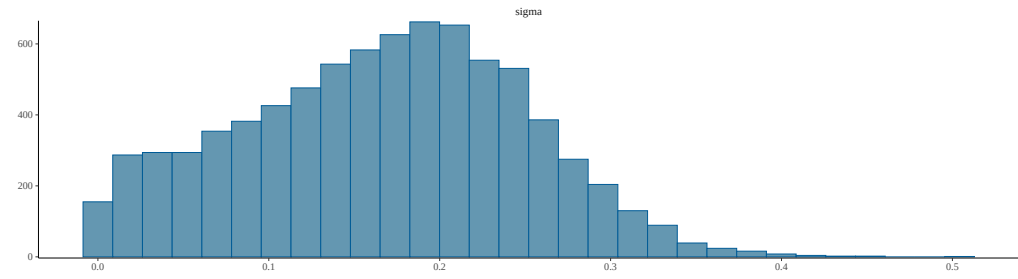
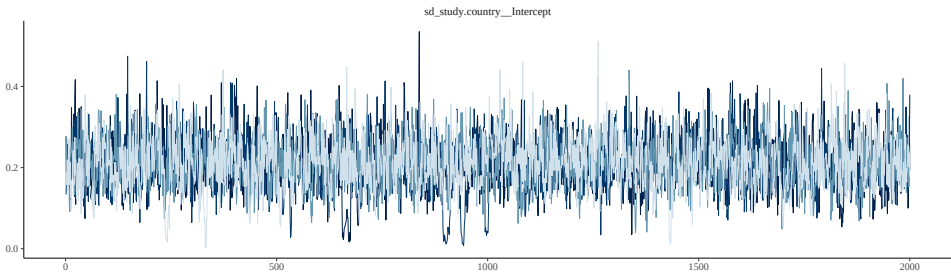
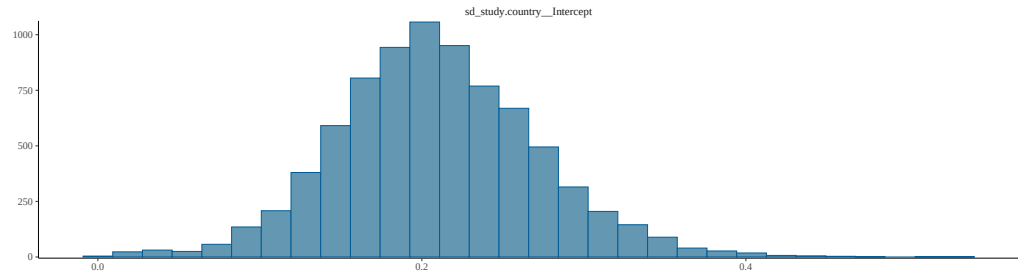
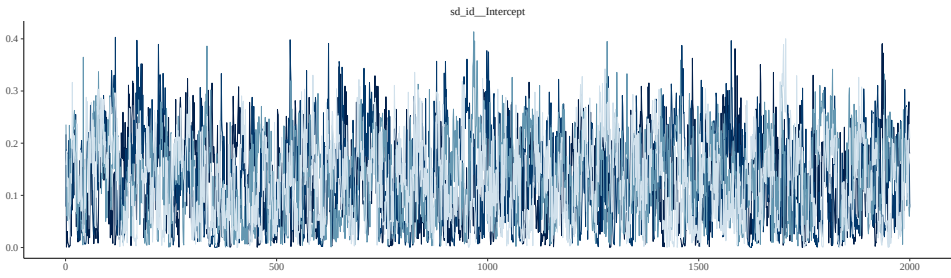
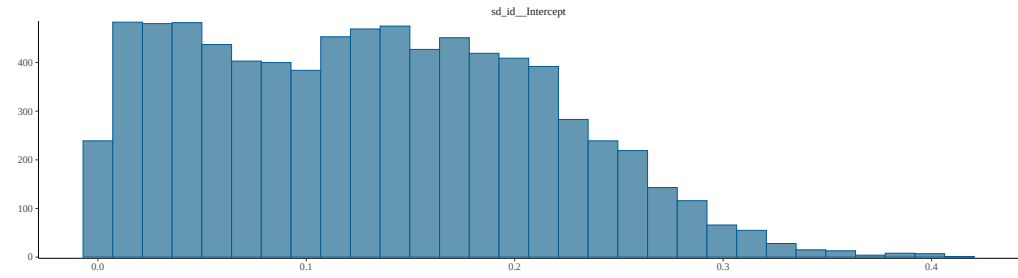
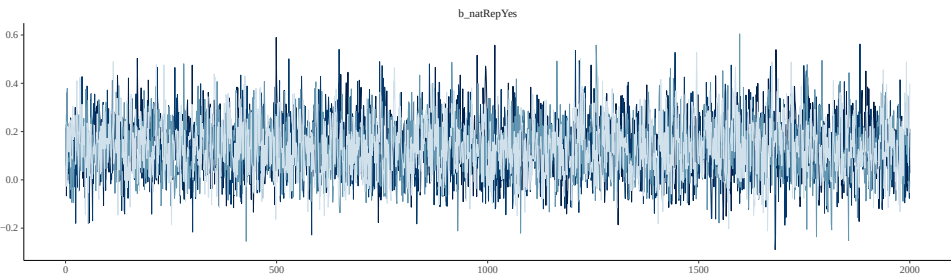
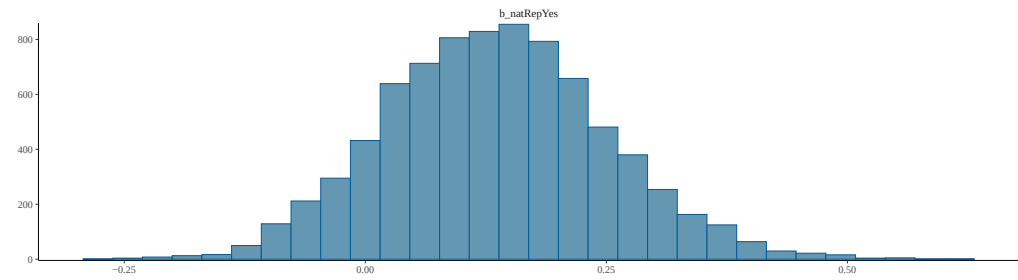
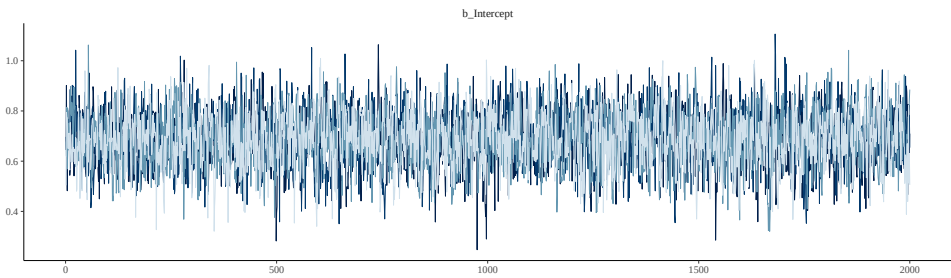
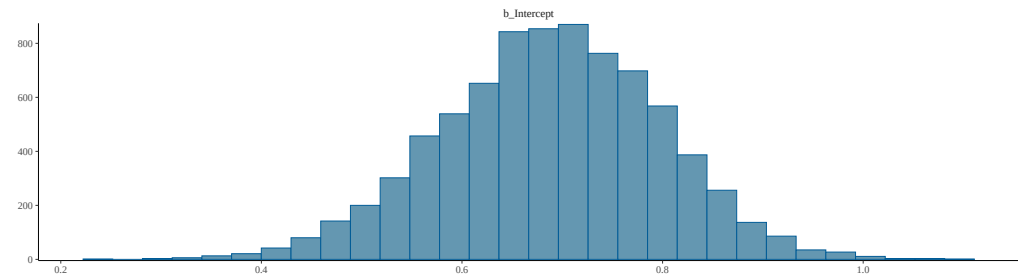
Summarize and check

```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + natRep + (1 | study.country) + (1 | id)
## Data: bactModelDF (Number of observations: 99)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 99)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 0.13 0.08 0.01 0.29 1.00 918 2462
##
## ~study.country (Number of levels: 33)
```

```

##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)      0.21      0.06      0.09      0.34 1.00      2313      1808
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      0.69      0.11      0.47      0.90 1.00      3427      4499
## natRepYes      0.13      0.11     -0.08      0.37 1.00      4141      5003
##
## Further Distributional Parameters:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      0.17      0.08      0.01      0.32 1.00      911      1726
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

```

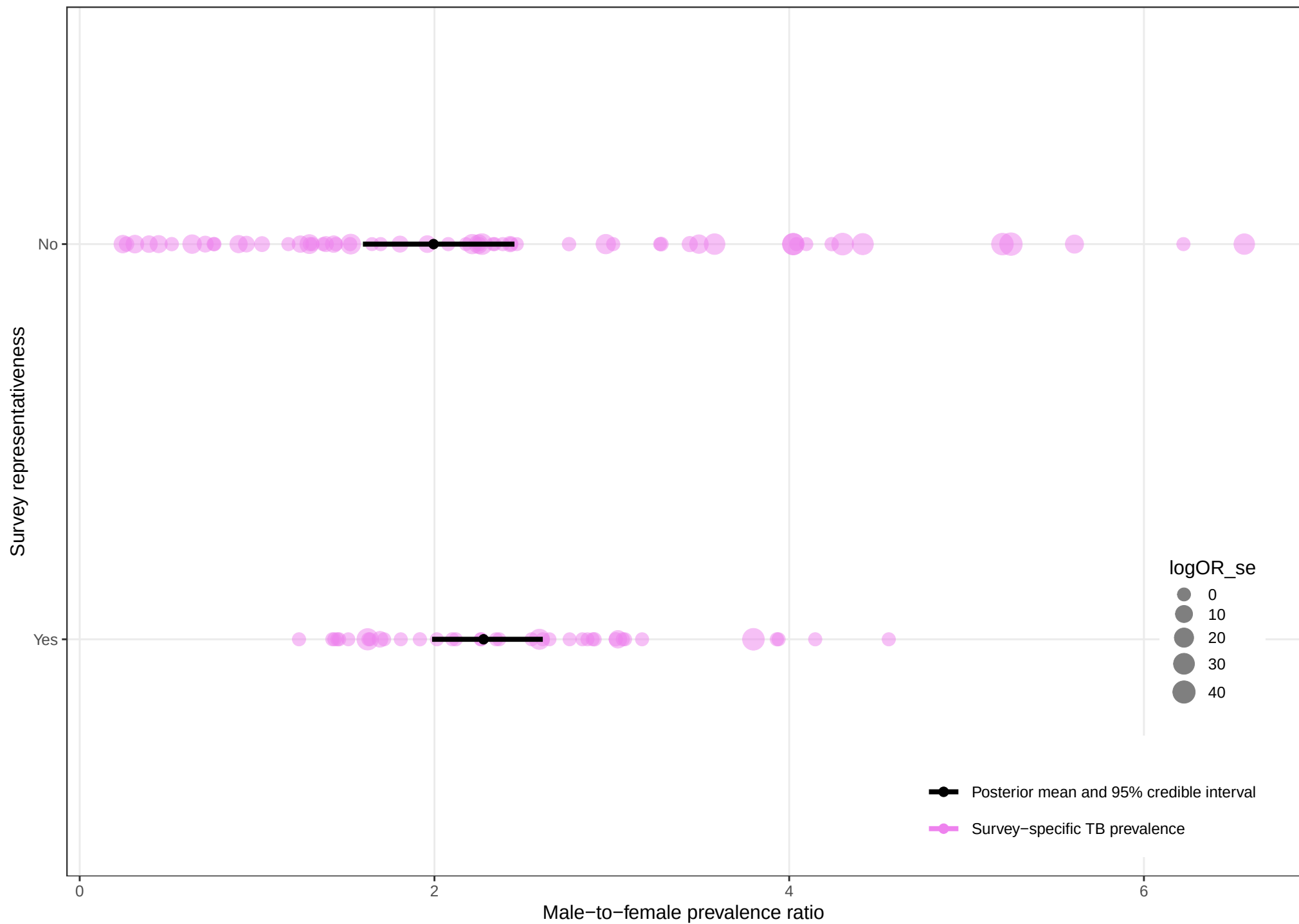



Prediction

Main estimate

```
natEffectModEst %>%  
  group_by(natRep) %>%  
  mean_qi(mfRatio) %>%  
  select(natRep, mfRatio, .lower, .upper)
```

```
## # A tibble: 2 x 4  
##   natRep mfRatio .lower .upper  
##   <chr>   <dbl> <dbl> <dbl>  
## 1 No      2.00    1.60  2.45  
## 2 Yes     2.28    1.99  2.61
```



Model 6: Study quality (High vs. not high)

Model specification

```
biasEffectMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + bias + (1 | study.country) + (1 | id),
  data = bactModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd),
  family = gaussian(),
  control = list(adapt_delta = 0.99),
  chains = 4,
  cores = 4,
  iter = 4000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##      ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

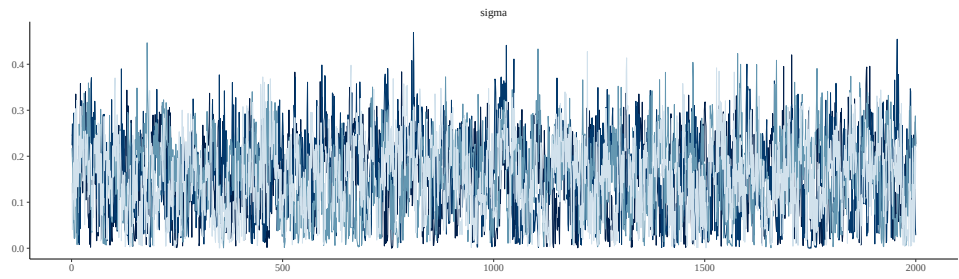
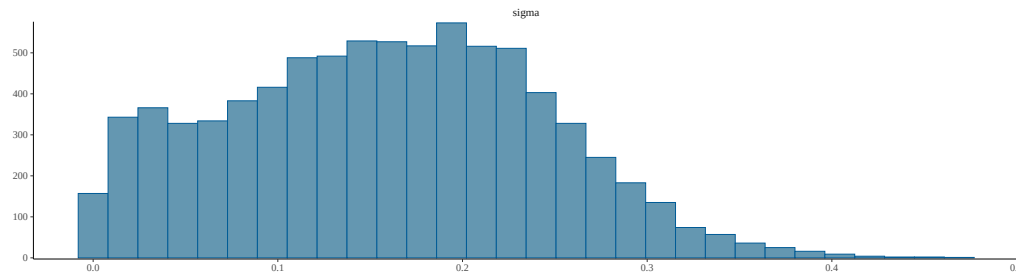
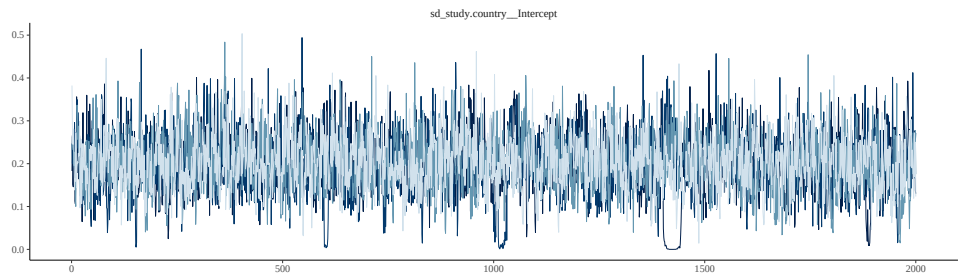
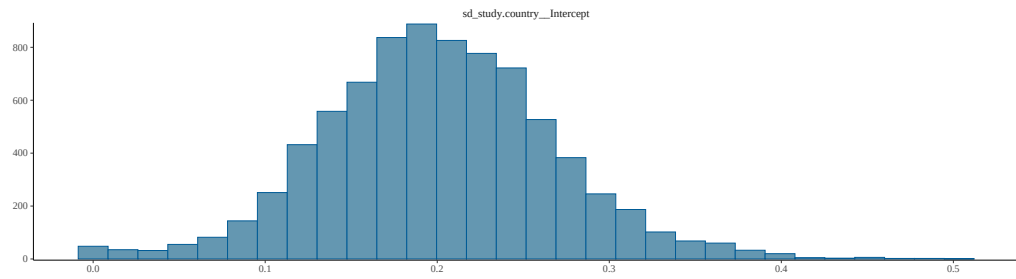
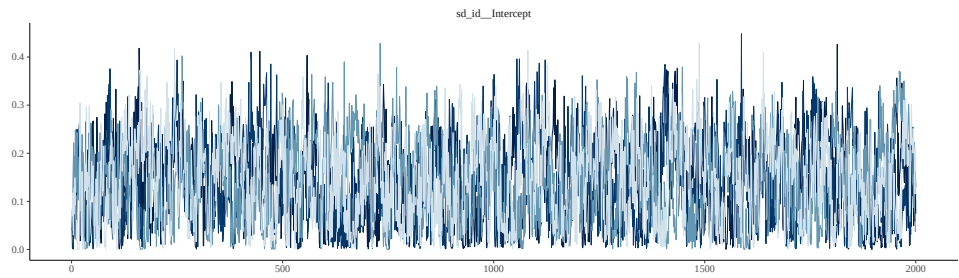
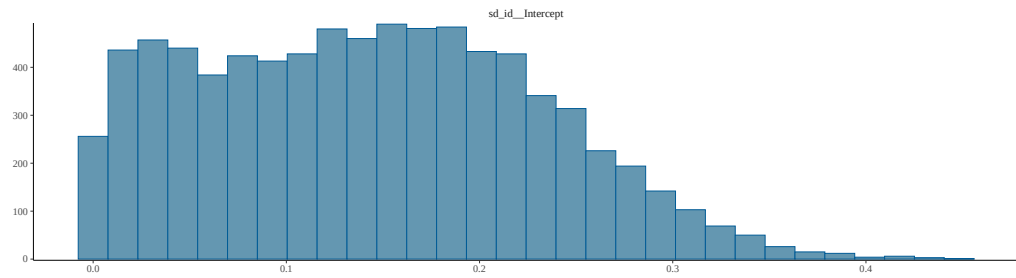
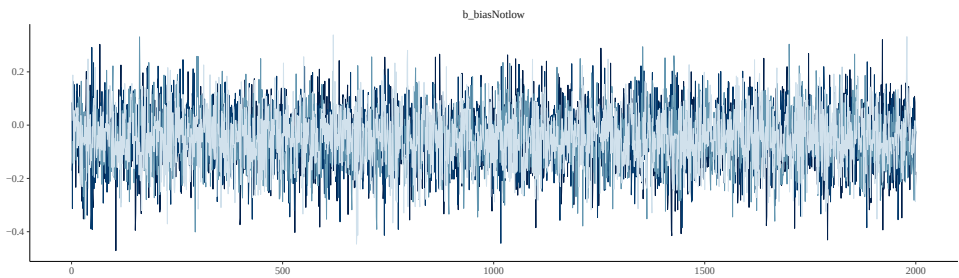
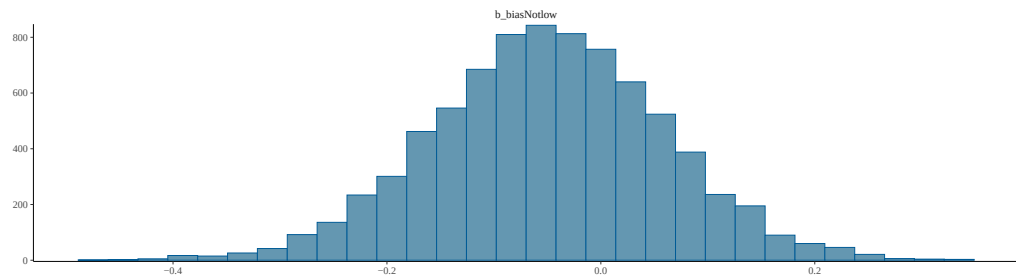
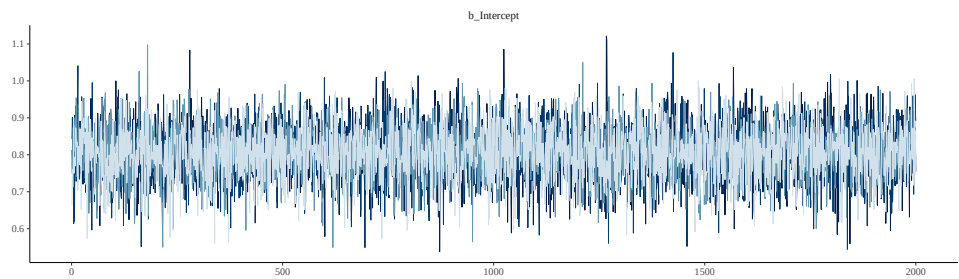
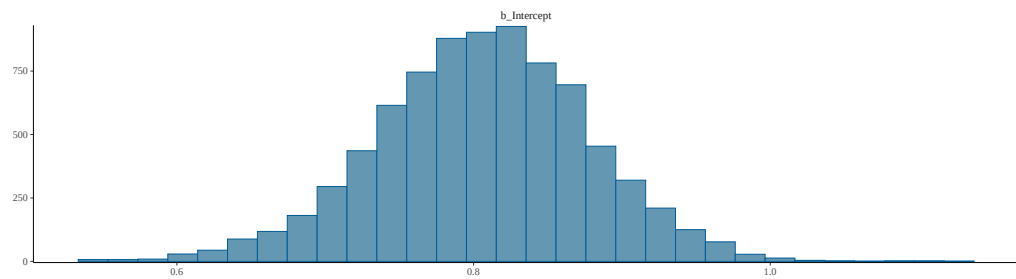
Summarize and check

```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + bias + (1 | study.country) + (1 | id)
## Data: bactModelDF (Number of observations: 99)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 99)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.14    0.09    0.01    0.31 1.00    723    1786
##
## ~study.country (Number of levels: 33)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.20    0.07    0.07    0.34 1.00   1624    1018
##
```

```

## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      0.81      0.07   0.66   0.94 1.00    6912    6246
## biasNotlow    -0.05      0.11  -0.26   0.16 1.00    6602    6548
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      0.16      0.08   0.01   0.32 1.00     913    2571
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

```



Chain
1
2
3
4

Prediction

Main estimate

```
biasEffectModEst %>%  
  group_by(bias) %>%  
  mean_qi(mfRatio) %>%  
  select(bias, mfRatio, .lower, .upper)
```

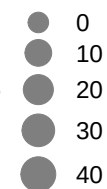
```
## # A tibble: 2 x 4  
##   bias      mfRatio .lower .upper  
##   <chr>      <dbl> <dbl> <dbl>  
## 1 Low          2.24  1.93  2.57  
## 2 Not low      2.14  1.76  2.60
```

Low probability of survey bias vs. not low probability of survey bias

Low

Not low

logOR_se



Posterior mean and 95% credible interval

Survey-specific TB prevalence

0

2

4

6

Male-to-female prevalence ratio

Model 7: Symptom screen vs. no symptom screen

Model specification

```
sympEffectMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + symptomScreen + (1 | study.country) + (1 | id),
  data = bactModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd),
  family = gaussian(),
  control = list(adapt_delta = 0.99),
  chains = 4,
  cores = 4,
  iter = 4000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##      ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

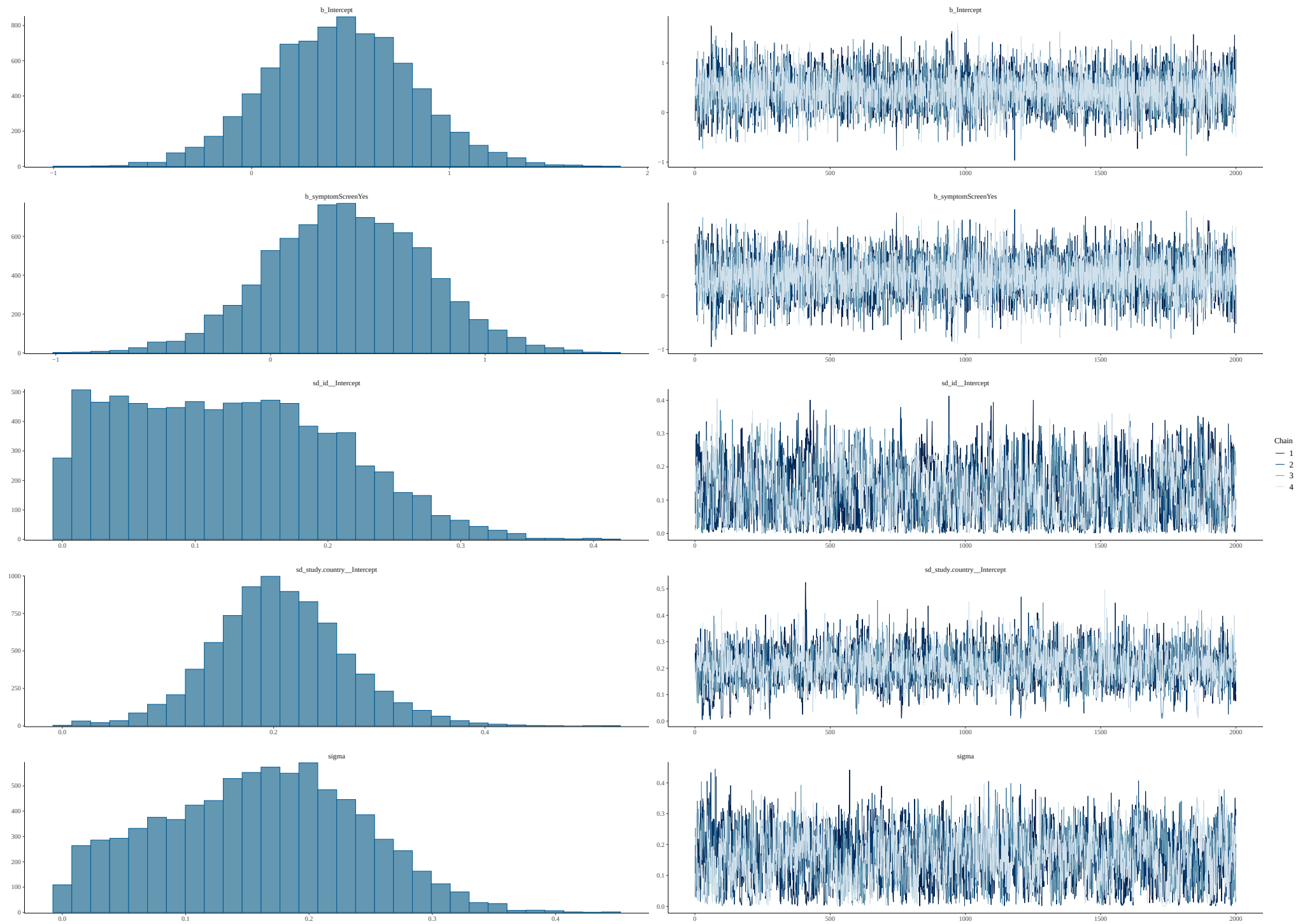
Summarize and check

```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + symptomScreen + (1 | study.country) + (1 | id)
## Data: bactModelDF (Number of observations: 99)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 99)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.13    0.08    0.01    0.29 1.00    922    2531
##
## ~study.country (Number of levels: 33)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.21    0.06    0.08    0.34 1.00   2559    2331
##
```

```

## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept           0.44      0.36  -0.27    1.16 1.00      7397      5472
## symptomScreenYes     0.36      0.37  -0.37    1.08 1.00      7803      5527
##
## Further Distributional Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      0.16      0.08    0.01    0.31 1.01       895      1947
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

```

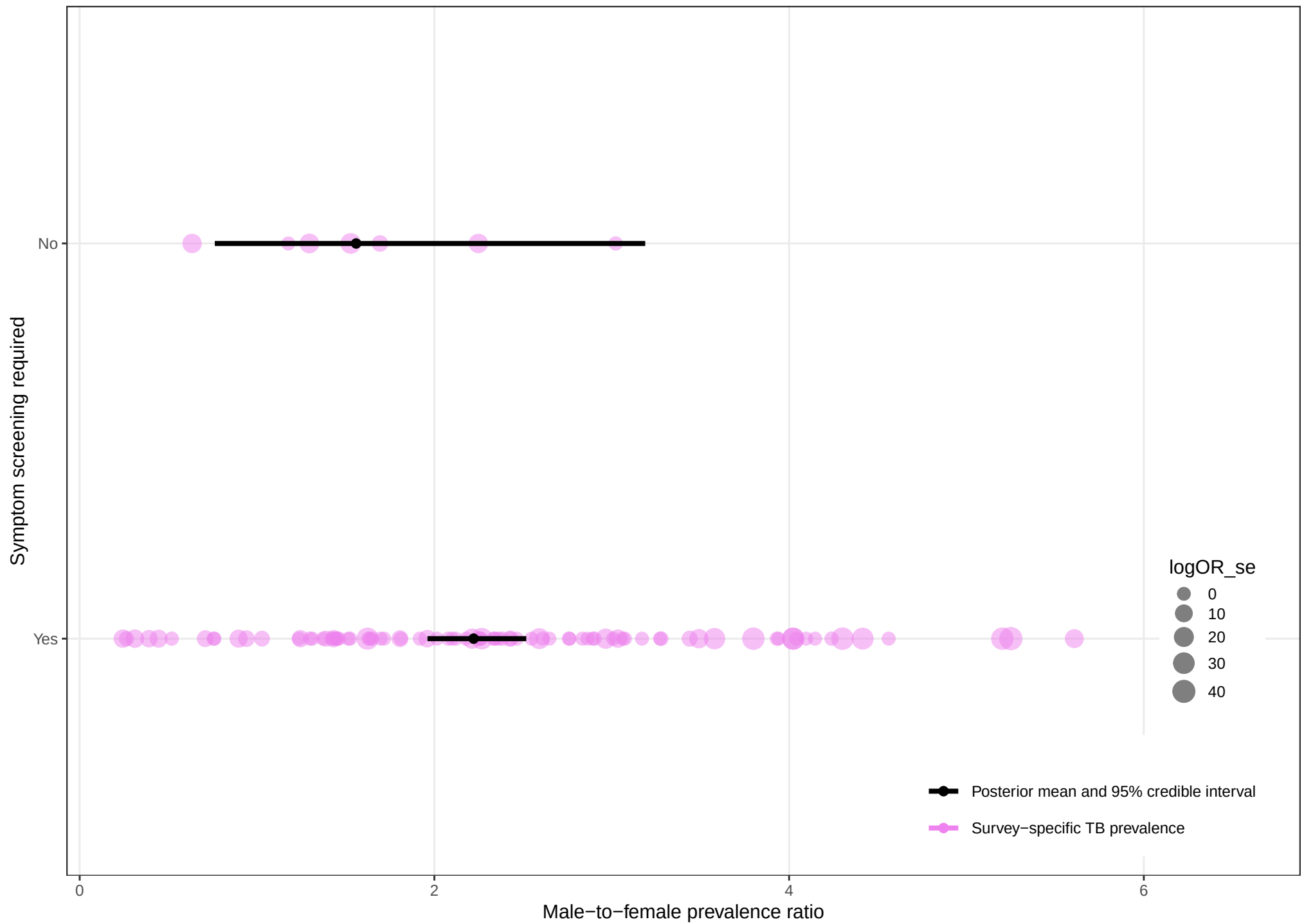


Prediction

Main estimate

```
sympEffectModEst %>%  
  group_by(symptomScreen) %>%  
  mean_qi(mfRatio) %>%  
  select(symptomScreen, mfRatio, .lower, .upper)
```

```
## # A tibble: 2 x 4  
##   symptomScreen mfRatio .lower .upper  
##   <chr>         <dbl>  <dbl>  <dbl>  
## 1 No           1.66  0.762   3.19  
## 2 Yes          2.23  1.96    2.52
```



Smear positive analysis

```
## [1] "distribution of samples for calculation:"
##
## adj.prev100k.ci.smear.positive.tb      n.smear.positive.tb
##                29                      27
##                none      prev100k.ci.smear.positive.tb
##                43                      7
##      prev100k.smear.positive.tb
##                1
```

Smear positive descriptive statistics

```
## [1] "64 unique surveys with sex-stratified smear positive TB"

## [1] "58 unique publications"

## [1] "63 studies report participant numbers totaling 2825955 participants"

## [1] "45.63% of participants were male"
```

Model X: Smear positive main effect

Model specification

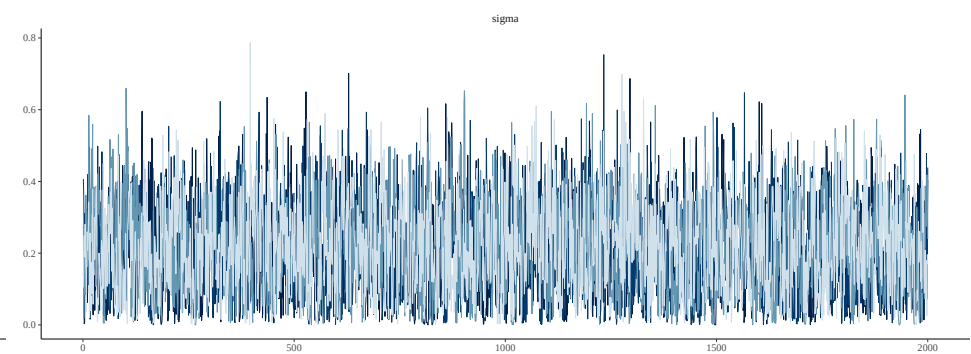
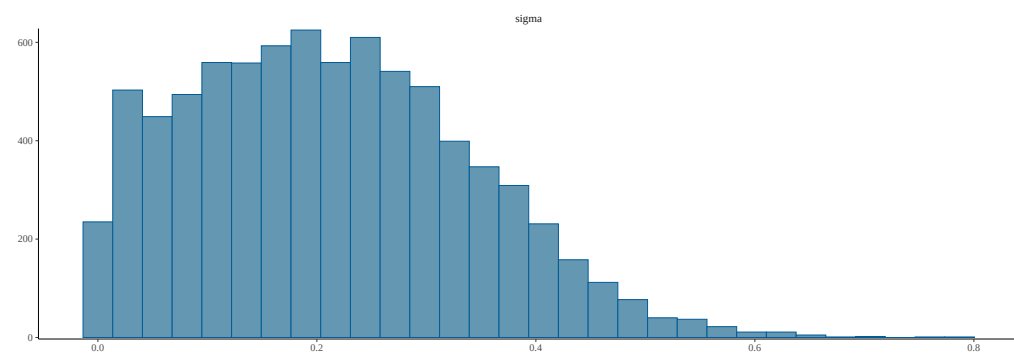
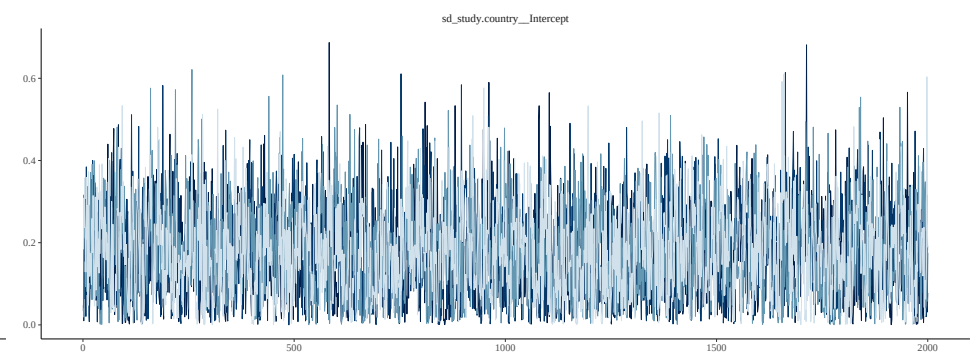
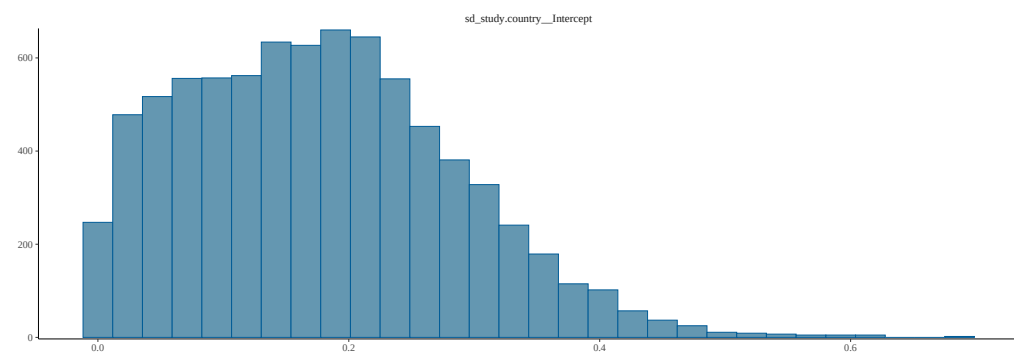
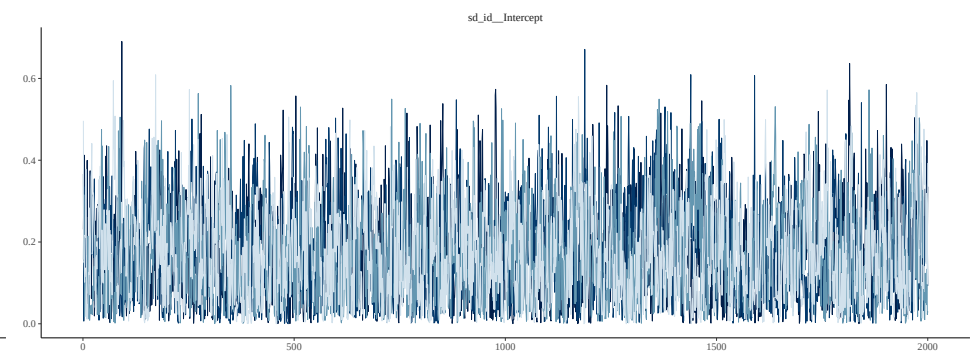
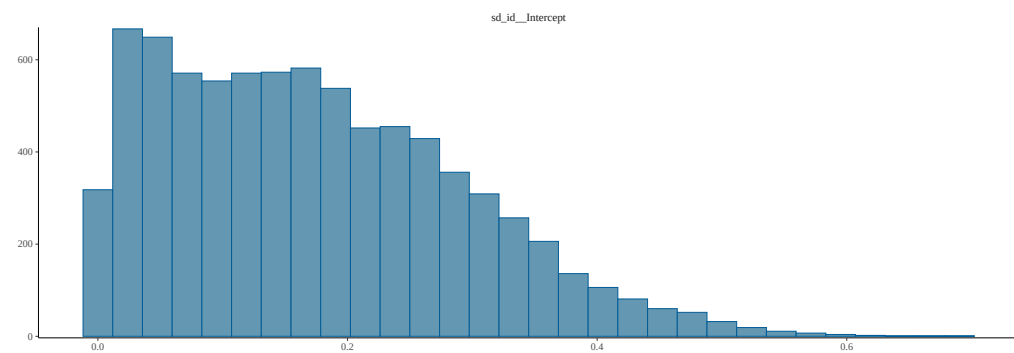
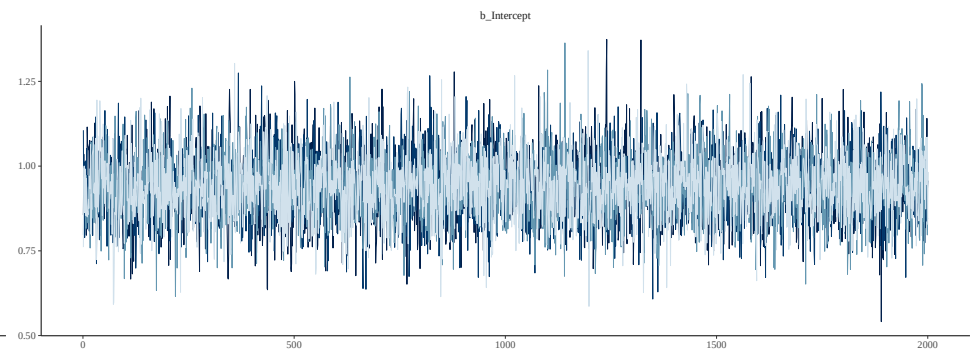
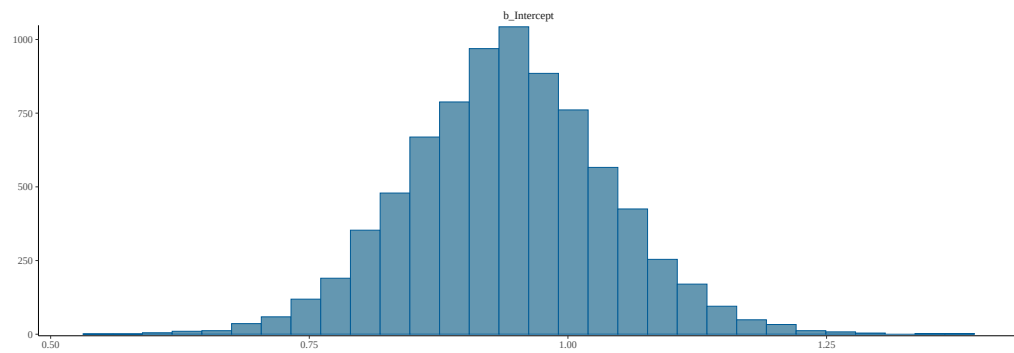
```
mainEffectSmrMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + (1 | study.country) + (1|id),
  data = smrModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    # prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd),
  family = gaussian(),
  control = list(adapt_delta = 0.99),
  chains = 4,
  cores = 4,
  iter = 4000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
```

```
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##      ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

Summarize and check

```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + (1 | study.country) + (1 | id)
## Data: smrModelDF (Number of observations: 63)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 63)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.17     0.12    0.01    0.44 1.00    1913    4061
##
## ~study.country (Number of levels: 26)
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.18     0.11    0.01    0.40 1.00    2410    3668
##
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    0.94     0.10    0.75    1.14 1.00     7926    6175
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma    0.21     0.13    0.01    0.48 1.00     2727    3126
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```



Chain
— 1
— 2
— 3
— 4

Prediction

Main estimate


```
mainEffectSmrModEst %>% ungroup %>% mean_qi(mfRatio)
```

```
## # A tibble: 1 x 6
##   mfRatio .lower .upper .width .point .interval
##   <dbl>   <dbl>   <dbl>   <dbl> <chr>   <chr>
## 1     2.58     2.13     3.11     0.95 mean     qi
```

Model 2: Smear positive effect by world region model

Model specification

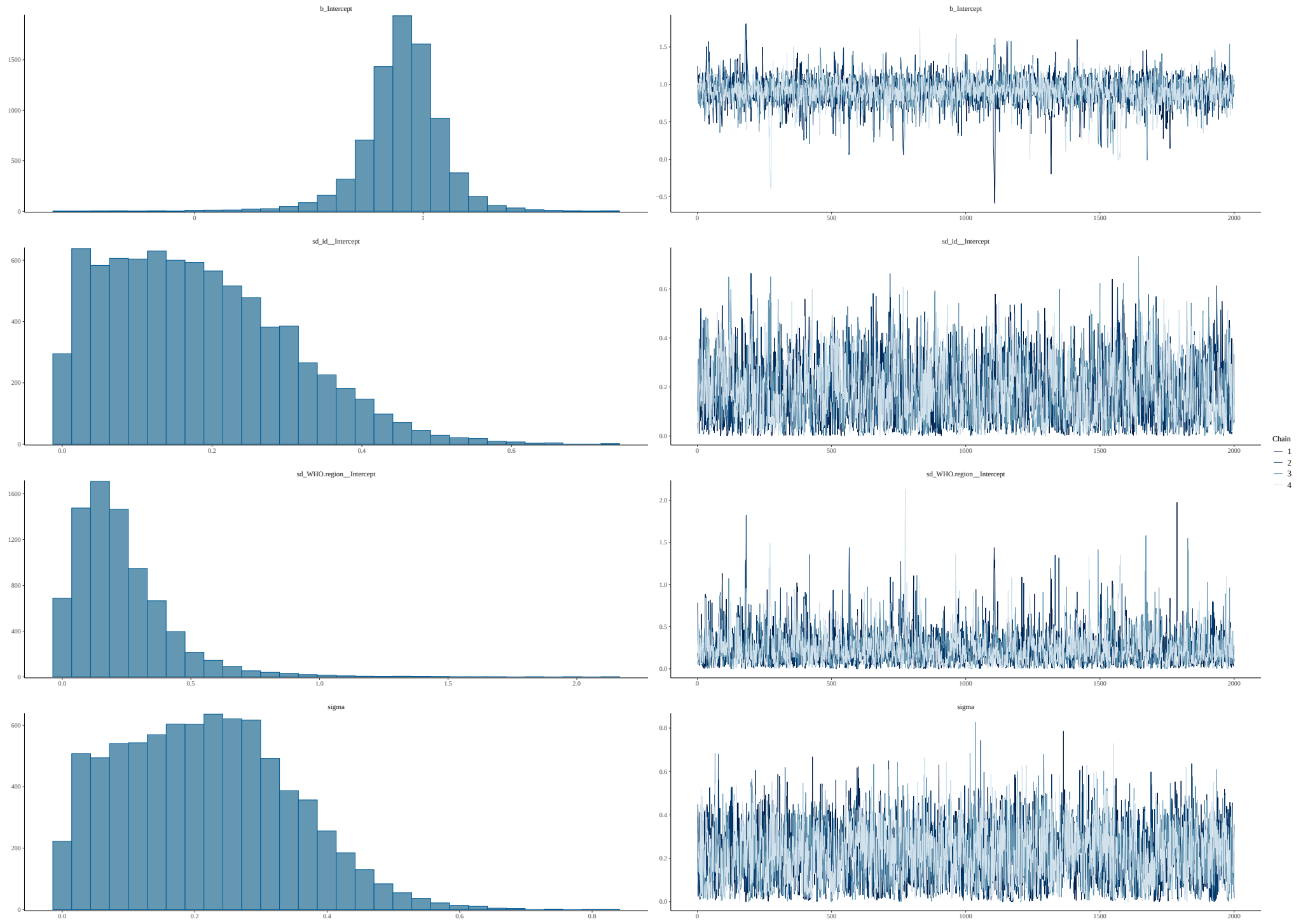
```
regionEffectSmrMod <- brms::brm(formula = logOR | se(logOR_se, sigma=TRUE) ~
  1 + (1|WHO.region) + (1|id),
  data = smrModelDF,
  prior = prior(student_t(7, 0, 1.5), class = Intercept) +
    # prior(normal(0, 10), class = b) +
    prior(exponential(2), class = sd),
  family = gaussian(),
  control = list(adapt_delta = 0.99),
  chains = 4,
  cores = 4,
  iter = 4000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' f
## #include <cmath>
##           ~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

Summarize and check

```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: logOR | se(logOR_se, sigma = TRUE) ~ 1 + (1 | WHO.region) + (1 | id)
## Data: smrModelDF (Number of observations: 63)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
```

```
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 63)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.18    0.12    0.01    0.44 1.00    1751    3852
##
## ~WHO.region (Number of levels: 5)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.23    0.19    0.01    0.70 1.00    2229    3047
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    0.91    0.18    0.51    1.22 1.00    3415    2674
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma    0.22    0.13    0.01    0.48 1.00    2390    2688
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```



Prediction

Main estimate

```
regionEffectSmrModEst %>% group_by(WHO.region) %>% mean_qi(mfRatio)
```

```
## # A tibble: 5 x 7
```

WHO.region	mfRatio	.lower	.upper	.width	.point	.interval
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1 Africa region	2.29	1.78	2.84	0.95	mean	qi
2 Americas region	2.54	1.14	4.36	0.95	mean	qi
3 Eastern Mediterranean region	2.39	1.56	3.29	0.95	mean	qi
4 South-East Asia region	2.95	2.17	4.03	0.95	mean	qi
5 Western Pacific region	2.69	2.07	3.49	0.95	mean	qi

